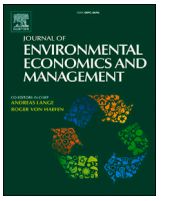
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Beyond the canopy: How satellite data detection thresholds influence policy evaluation and deforestation behavior☆

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ABSTRACT

Satellite data is essential for enforcing and evaluating environmental policy, but technological limitations of monitoring systems can create perverse incentives and bias impact assessment. This study examines how detection thresholds in satellite monitoring systems affect both the implementation and evaluation of forest conservation policies. We identify three key mechanisms: a measurement issue, where datasets with larger minimum detection thresholds systematically miss small-scale deforestation; a loophole effect, where policy only reduces detectable, large-scale deforestation; and strategic adaptation, where regulated agents adjust behavior to exploit known detection thresholds, substituting from large- to small-scale deforestation. Studying Brazil's 2008 municipal Blacklisting policy, we find that the government's primary monitoring system, which does not report patches below 6.25 hectares, overestimates policy effectiveness by a third compared to datasets with smaller minimum detection thresholds. When measured with those datasets, blacklisting reduced deforestation by 31.2 % from baseline—substantially less than the 47.6 % reduction suggested by government data. Average clearing size declined by 28.9 %, with significant increases in patches below detection thresholds, reflecting both undetected and strategically fragmented activity. Our analysis reveals a critical challenge for environmental governance: as monitoring systems improve, so too do evasion strategies, requiring close attention to how technology shapes observed outcomes and on-the-ground incentives.

1. Introduction

Deforestation in the Brazilian Amazon has long been recognized as a critical driver of climate change, contributing significantly to global carbon emissions and the loss of biodiversity. The intricate balance between the Earth's ecosystems is dependent on the stability of these forests, making the implementation of effective policies to combat deforestation an urgent priority. In order to design and assess the efficacy of these policies, accurate measurement of deforestation rates is fundamental. As satellite data becomes increasingly crucial for measuring and monitoring deforestation (Fuller, 2006), the choice of data source–particularly its detection threshold or mapping unit–emerges as a key factor in both policy evaluation and implementation (Ferraro, 2009; Avelino et al., 2016; Jain, 2020; Auffhammer et al., 2020; Alix-Garcia and Millimet, 2023; Proctor et al., 2023; Garcia and Heilmayr, 2024). Datasets with smaller minimum mapping units can detect smaller deforestation patches that are often missed by coarser datasets, which have traditionally

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been used for monitoring and enforcing deforestation policies. The implications of this data choice are profound: it affects how we measure the success of conservation policies and how agents on the ground adapt their behavior to avoid detection.

This paper examines how the technical limitations of monitoring systems can lead to measurement error, loopholes in detection, and strategic adaptation by regulated agents—ultimately undermining policy impact and complicating impact evaluation. We identify three interrelated mechanisms through which monitoring technology influences environmental policy outcomes. First, a general measurement issue arises when datasets with different detection thresholds capture varying levels of environmental damage. Datasets constrained by a higher minimum mapping threshold may overlook deforestation in smaller patches, leading to systematic underestimation of total deforestation. This has important implications for how we evaluate conservation policies: as these coarser datasets fail to register smaller-scale clearing, they may understate the extent of the forest loss issue, and change estimates of policy effectiveness by establishing noisy baseline levels of deforestation.

Second, we identify a loophole effect[1] that arises directly from these detection thresholds. Because some satellite-based deforestation datasets only record patches above a given size, deforestation that occurs below that threshold–even if ongoing and substantial–remains invisible to the monitoring system. As a result, enforcement and evaluation may miss a large share of total forest loss. Importantly, this loophole exists even in the absence of deforesters actively substituting into smaller patches: pre-existing small-scale deforestation that continues unchanged after policy implementation will appear as a policy success in the data. The satellite system records reduced large-scale deforestation (which the policy may have genuinely curtailed) while remaining blind to unchanged small-scale activity. This measurement limitation alone can lead to systematic overestimation of policy effectiveness, as the apparent reduction in total recorded deforestation conflates genuine policy impact with the invisibility of small-scale forest loss.

Third, and more pervasively, we consider the possibility of strategic adaptation by regulated agents. When deforesters are aware of satellite detection constraints, they may deliberately adjust their behavior to exploit these monitoring gaps. In the context of satellite-based enforcement, this typically manifests as the intentional fragmentation of deforestation activities into smaller patches that fall below minimum detectable thresholds—effectively substituting large-patch deforestation (which can be detected and penalized) for small-patch deforestation (which cannot). This strategic response creates a compounding problem for policy evaluation. Unlike the loophole effect described above, where small-scale activity simply remains constant and invisible, strategic adaptation actively increases small-scale deforestation in response to policy implementation. The result is a double distortion: satellite data shows reduced large-patch activity (suggesting policy success) while missing both pre-existing small-scale activity and new small-scale activity generated by the policy itself. This behavioral response not only undermines enforcement effectiveness by making violations harder to detect and prosecute, but also leads to even greater overestimation of policy impact.

We study these dynamics in the context of Brazil's deforestation monitoring system, where policy implementation and evaluation have historically relied on the government's Program for the Calculation of Deforestation in the Legal Amazon (Programa de Cálculo do Desflorestamento da Amazônia Legal–PRODES) dataset. While PRODES provides comprehensive coverage, it only publicly reports deforestation patches larger than 6.25 hectares. To assess the implications of this threshold, we compare PRODES with datasets like MapBiomas, which can detect forest loss in patches as small as approximately 0.5 hectares (MapBiomas, 2023). Both datasets rely on the same 30-meter Landsat imagery, so differences in detection stem not from pixel resolution but from minimum mapping units. This comparison allows us to isolate the extent to which policy evaluations are sensitive to the detection capacity of the underlying data.

This behavioral adaptation to satellite-based monitoring is not merely theoretical. In a recent interview, the former Director of Environmental Protection at the Brazilian Institute of Environment and Renewable Natural Resources (Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis–IBAMA) describes how in 2014, an illegal deforestation operation in the Amazon had developed sophisticated methods to evade detection. According to Evaristo (2023), the leader of this operation "had created a new deforestation model specifically designed to deceive the DETER satellite".[2] The operation organized dozens of small camps, each employing eight chainsaw operators who would clear precisely 2.72 hectares[3] per day–intentionally staying below PRODES's 6.25-hectare detection threshold.[4] The sophistication of this evasion strategy extended to hiring GIS specialists to monitor satellite detection and relocate operations if discovered. This example illustrates how deforestation agents actively adapt their behavior in response to known limitations of satellite monitoring systems, underscoring the importance of understanding both the technical constraints of different datasets and the behavioral responses they induce.

Our analysis reveals that PRODES systematically underestimates deforestation rates compared to datasets with smaller minimum detection thresholds, capturing only about three-quarters of the total deforestation detected by MapBiomas. We then assess how satellite data choice shapes both the implementation and evaluation of deforestation policies, focusing on the Blacklisting of municipalities. The Blacklisting policy, or Priority Municipalities (Municípios Prioritários–MPs), initiated in 2008 under Brazil's Action Plan

[1] We are grateful to an anonymous reviewer and the Editor for suggesting this language to describe this mechanism.

[2] Real-Time Deforestation Detection System (Detecção do Desmatamento em Tempo Real–DETER) is an alert-based system developed by Brazil to support rapid environmental enforcement by identifying recent deforestation activity.

[3] 2.72 ha (1 alqueire) is a common and culturally salient land unit in Brazil. An alqueire is a traditional unit of measurement in Portuguese, and is widely used in rural Brazil to measure units of land.

[4] While PRODES uses higher-resolution Landsat imagery, it only detects and reports clearings ≥6.25 hectares. DETER originally had thresholds of 25 ha until 2015; since then (with DETER-B), it issues alerts for areas as small as approximately 3 ha, though publicly shared alerts are filtered to ≥6.25 hectares to match PRODES standards. Because this study focuses on 2004–2012, when we speak about DETER threshold, we will be referring to the 25 hectare detection threshold. These thresholds reflect minimum mapping units rather than true detection capabilities, which vary with cloud cover, forest type, and revisit frequency. Both systems remain vulnerable to evasion strategies that exploit operational constraints.

for the Prevention and Control of Deforestation in the Legal Amazon (Plano de Ação para a Prevenção e Controle do Desmatamento na Amazônia Legal–PPCDAm), targets municipalities with persistently high deforestation rates by imposing stricter environmental monitoring, agricultural credit restrictions, and sanctions. The goal of the policy is to incentivize reductions in deforestation by increasing its costs and strengthening enforcement. Blacklisting serves as an ideal case study because it relied on the PRODES dataset not only for implementation–by identifying municipalities with high deforestation–but also for evaluating policy success. At the same time, the program's enhanced monitoring and sanctions likely altered the incentives of deforesters, creating conditions where agents on the ground were highly motivated to evade detection, especially by exploiting the known limitations of the monitoring systems in use.

To analyze these dynamics, we exploit the staggered rollout of the blacklist across municipalities, using synthetic difference-in-differences regressions that correct for selection into treatment and non-parallel pre-trends. Our findings show that PRODES consistently overestimates the effects of blacklisting, with estimated impacts roughly one-third larger than those obtained using MapBiomas. Using PRODES data, we estimate that blacklisting reduced annual deforestation by 0.502 percentage points, or 47.6 % relative to baseline levels in treated municipalities. By contrast, MapBiomas indicates a smaller reduction of 0.398 percentage points, or 31.2 %. This sizable gap in estimated effects suggests that relying on datasets with larger minimum detection thresholds can significantly exaggerate the apparent success of anti-deforestation policies.

This overestimation arises from two interrelated mechanisms. First, a loophole effect emerges from PRODES's detection threshold: because the system only reports patches larger than 6.25 hectares, any deforestation occurring in smaller patches goes unreported, even if policy has no behavioral effect. We find clear evidence of this: after blacklisting, the share of deforestation occurring in large patches (e.g., 50–100 ha and 200–1000 ha) declines sharply, while deforestation in small patches increases, particularly in the 0–6.25 ha range. Second, there is evidence of strategic adaptation. Following blacklisting, we document a 28.9 % decline in the mean patch area of deforestation, driven by increases in the number and share of small patches–suggesting that agents deliberately adjusted the scale of their clearing to evade detection thresholds. The strongest increases occur in the share of deforestation accounted for by patches smaller than 6.25 hectares, and also in those between 6.25 and 25 hectares–sizes that are below the operational detection range of Brazil's near-real-time monitoring system (DETER), though still technically captured by PRODES.

These findings highlight how agents exploit both measurement gaps and monitoring constraints, and how these behaviors can compound to bias evaluations. If policy evaluation relies on satellite datasets that miss small-scale clearing, it may overstate policy effectiveness—not only because of missed detection, but also because enforcement itself may induce behavioral responses such as only reducing large scale deforestation or even substituting large scale for small-scale deforestation, which makes illegal behavior harder to observe. Recognizing the implications of these mechanisms is critical for both program design and evaluation. Underestimating deforestation can lead to the premature celebration of ineffective policies or the continued endorsement of programs that are being strategically undermined. It also raises the possibility that policymakers or stakeholders may selectively rely on more favorable datasets when evaluating outcomes. Our results underscore the need to use satellite data with smaller minimum detection thresholds and local calibration to accurately monitor forest change and assess the true impacts of environmental interventions.

Our paper makes several key contributions to the literature on deforestation policy evaluation, environmental economics, and the use of remote sensing data. First, our paper expands on the literature examining strategic adaptation to policy and regulation by focusing on how deforestation agents modify their behavior in response to increased monitoring. Similar to the findings of Olken and Barron (2009), who documented how Indonesian truck drivers adapted to avoid extortion, we show that deforestation agents in Brazil strategically reduce the size of deforestation patches to evade detection. This behavior mirrors the dynamics in Duflo et al. (2013), where firms manipulated environmental reports in response to audits. In our context, intensified monitoring in blacklisted municipalities leads to smaller, harder-to-detect deforestation patches, complicating enforcement. This underscores the broader theme that policy interventions, while well-intentioned, can produce unintended strategic behaviors that weaken their effectiveness. This aligns with findings that environmental offsets can sometimes encourage harmful behaviors (Kotchen, 2006, 2009). In the case of Brazil's Blacklisting policy, we demonstrate that stricter enforcement prompts strategic deforestation evasion, paralleling unintended effects observed in policies such as residential energy consumption (Jacobsen et al., 2012) and building energy codes (Kotchen and Grant, 2011). These findings underscore the importance of anticipating behavioral responses when designing environmental policies.

Second, we add to the literature on the impact of data resolution on policy effectiveness. Prior work by Alix-Garcia and Millimet (2023), Proctor et al. (2023), and Garcia and Heilmayr (2024) has demonstrated how varying resolutions can lead to different assessments of policy impact. In the Brazilian Amazon context, Kalamandeen et al. (2018) document a pervasive rise in small-scale deforestation using Global Forest Change (GFC) data, showing that small clearings (<1 ha) increased by 34 % when comparing the periods 2001–2007 and 2008–2014, accounting for about 39 % of total Amazonian forest loss in the later period. While their analysis reveals these important patterns, it is descriptive in nature and does not identify the underlying causes of these changes nor does it connect them to policy-relevant thresholds like PRODES's 6.25 ha or DETER's 25 ha cutoffs. Moreover, as Garcia and Heilmayr (2024) highlight, their use of GFC data has important limitations: it provides only binary, non-repeatable outcomes, meaning deforestation can be captured just once within each pixel. Our paper addresses these challenges by using PRODES alongside Mapbiomas data, which allows us to resolve many of the concerns around binary non-repeatable outcomes.[5] Using the Blacklisting program as a significant shock to enforcement, we demonstrate not only how the choice of satellite data affects perceptions of policy success, but also that

[5] Mapbiomas allows us to track both deforestation and reforestation within the same pixel over time, ensuring that the non-repeatability issue is not present in our data.

the rise in small-scale deforestation reflects strategic behavior by regulated agents. We go beyond issues of measurement error to examine how agents respond when monitoring systems are "blind" to smaller deforestation patches due to large minimum detection thresholds. Thus, our paper adds to the growing body of work on satellite data use by incorporating both behavioral responses and technical considerations around resolution, minimum detection thresholds and classification.

Third, our study advances the application of remote sensing data within environmental economics by implementing a highly scalable framework for analyzing deforestation at the pixel level across the Legal Amazon. The integration of Google Earth Engine (GEE) and the Landsat satellite archives has transformed the study of land-use change, as noted by Donaldson and Storeygard (2016), Straub et al. (2019), and Wulder et al. (2022). Building on these technological advancements, our analysis employs GEE in conjunction with Python and R-based landscape metric tools to process approximately 180 billion pixels, covering the full spatial extent of the Brazilian Amazon. We further adapt the FragStats methodology (McGarigal, 1995; Hesselbarth et al., 2019; Nowosad and Stepinski, 2019) within this big-data framework, allowing the computation of detailed patch-level statistics. This adaptation allows us to assess the size, distribution, and spatial characteristics of deforested areas before and after policy interventions with exceptional granularity. Our approach not only reveals deforestation patterns but also uncovers strategies of evasion by deforesting agents, demonstrating the significant policy insights gained from such data-intensive analysis. By tailoring our framework to process deforestation data from both MapBiomas and PRODES datasets, we provide a robust and replicable method for analyzing land-use changes across vast geographies.

Finally, we contribute to the growing body of work evaluating deforestation programs (Gibbs et al., 2015; Cisneros et al., 2015; Assunção and Rocha, 2019; Assunção et al., 2023) by providing a comprehensive analysis of Brazil's Blacklisting program. Building on prior evaluations by Cisneros et al. (2015), Assunção and Rocha (2019), and Assunção et al. (2023), we offer new insights into how this program influences deforestation outcomes across municipalities. By leveraging satellite data and advanced econometric techniques, our study deepens the understanding of the policy's effectiveness, while highlighting the importance of satellite-data detection thresholds in policy evaluation.

2. Satellite data and deforestation policy in Brazil

Forest governance and monitoring initiatives have significantly benefited from recent advancements in estimating fine-scale temporal and spatial deforestation rates using remotely sensed data (Fuller, 2006; Moffette et al., 2021). Platforms like Google Earth Engine (GEE) have revolutionized large-scale computing by co-locating data catalogs and processing capabilities, making remotely sensed data widely accessible (Gorelick et al., 2017). This has enabled consistent deforestation monitoring and estimates with global coverage, as demonstrated by initiatives like the Global Forest Watch (GFW) (Hansen et al., 2013; Global Forest Watch, 2022). In the case of Brazil, the Mapbiomas project has leveraged the entire Landsat archive to construct annual land-use classification maps (Souza et al., 2020).

Remotely sensed data offer several advantages for applied economics research, providing consistent, high-resolution, and long-term observations of phenomena (Donaldson and Storeygard, 2016). These data can facilitate the identification of causal relationships by providing quasi-experimental variation in exposure to environmental or policy changes (Jayachandran et al., 2017). However, establishing causality using satellite data can be challenging, as observational data may not provide sufficient control over confounding factors (Andam et al., 2008; Ferraro, 2009). Careful attention must be paid to the choice of data sources and methodological approaches to avoid biased policy evaluation results (Jain, 2020; Alix-Garcia and Millimet, 2023; Proctor et al., 2023; Garcia and Heilmayr, 2024).

Satellite data has played a crucial role in monitoring and enforcing deforestation policies in Brazil. The country's Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), implemented in 2004, relied heavily on satellite data to track deforestation rates (Assunção et al., 2015, 2023). The Program for the Calculation of Deforestation in the Legal Amazon (PRODES) dataset, developed by the Brazilian government, is the primary tool used to monitor deforestation in the Amazon (Instituto Nacional de Pesquisas Espaciais, 2021b).

We begin by presenting two widely used sources of deforestation data in Brazil: PRODES and MapBiomas. Although both systems use the same 30-meter Landsat imagery, they differ significantly in their detection thresholds. PRODES, the primary government monitoring system, employs detection algorithms with a minimum mapping unit of 6.25 hectares, meaning it cannot capture smaller clearing events. In contrast, MapBiomas's detection algorithms take full advantage of the underlying Landsat data resolution, allowing identification of deforestation patches as small as 0.09 hectares (one Landsat pixel). This difference in detection capability has important implications for both policy implementation and evaluation. We compile a municipality-year panel dataset for 773 municipalities in the Brazilian Amazon from 2004 to 2012, measuring annual deforestation using both of these satellite-based datasets. We then go on to analyze how the choice of deforestation data can impact the measurement of deforestation, as well as the evaluation and implementation of policies aimed at reducing deforestation in the region.

2.1. Different sources of deforestation data

PRODES: PRODES, maintained by Brazil's National Institute for Space Research (Instituto Nacional de Pesquisas Espaciais–INPE), is the official monitoring system for deforestation in the Brazilian Amazon. It focuses on clear-cut deforestation, utilizing satellite imagery from Landsat and other sensors to detect forest losses greater than 6.25 hectares in primary forest areas (Instituto Nacional de Pesquisas Espaciais, 2021b,a). PRODES reports annual deforestation rates across Brazilian municipalities since 1988 and has been used to generate the official government statistics on annual deforestation rates in the Brazilian Legal Amazon (Almeida et al., 2013). The PRODES system identifies clear-cut deforestation of only the old-growth forests, and once an area is identified as deforested, it

is not revisited in subsequent years. This means that the deforestation rate captured by PRODES is the annual increment of previous years. The mapping year (reference year) comprises August 1 of the prior year to July 31 of the reported year. PRODES uses a combination of medium spatial resolution images (20 to 30-m) to minimize cloud cover interference and ensure interoperability, but due to its minimum mapping unit, deforestation patches smaller than 6.25 hectares are not reported. As a result, smaller-scale forest losses–though potentially detectable in raw imagery–are excluded from official PRODES estimates (Rosa et al., 2012; Richards et al., 2017).

Mapbiomas: Mapbiomas is a non-governmental initiative that uses a combination of high-resolution satellite images and machine-learning algorithms to map land use and land-cover changes in Brazil, including deforestation (Souza et al., 2020). MapBiomas Collections (1-8) map Brazil's annual land cover and land use at a 30-meter resolution (based on Landsat data), with improved accuracy in biomes and new agricultural classes (Souza et al., 2020). Mapbiomas uses the new maps of biome limits produced by IBGE on a scale of 1:250,000 and is one of the most extensive land use classification projects available, estimating forest cover, secondary forest growth, and primary forest growth, as well as many other land-use classifications calibrated to the Brazilian reality. The minimum mapping unit of the Mapbiomas dataset is 0.09 hectares (the size of one 30×30 m pixel). However, MapBiomas applies a spatial filter to reduce classification error, which may exclude some noisily classified changes when they are below approximately 0.5 hectares.

A Note on Global Forest Watch (GFW): The Global Forest Change (GFC) dataset, developed by Hansen et al. (2013), provides annual estimates of tree-cover loss and gain based on Landsat and MODIS data. This dataset is disseminated through the Global Forest Watch (GFW) platform, an independent initiative led by the World Resources Institute (WRI), which offers public access to forest monitoring tools and global forest change visualizations (Global Forest Watch, 2022). The GFW dataset is based on the Hansen Global Forest Change v1.8, which shows 20 years (2000–2020) of tree-cover loss estimates worldwide (Hansen et al., 2013; Global Forest Watch, 2022). While Global Forest Watch (GFW) provides valuable global tree cover loss estimates, its calibration to worldwide data, rather than local Brazilian data, renders it less suitable for rigorous policy evaluation in the Brazilian Amazon. As highlighted by Nesha et al., 2021, externally generated remote sensing products often lack alignment with national classification systems and forest definitions. In contrast, datasets like MapBiomas and PRODES are calibrated to Brazil's legal framework and ecological biomes. GFW also displays significant volatility in its time series,[6] with sharp spikes and structural breaks—particularly after 2011 and 2015—caused by changes in its detection algorithms rather than real forest dynamics (Global Forest Watch, 2017, 2024). Notably, the GFW methodology was also revised in 2011, leading to visible discrepancies in the reported tree cover loss for 2011 and 2012 that are attributable to processing changes (Global Forest Watch, 2017, 2024). Given these limitations—and the significant computational burden of processing multiple datasets—we exclude GFW from our analysis.[7]

2.2. *Why these datasets might lead to issues in measuring deforestation and evaluating deforestation policy*

The PRODES and MapBiomas datasets employ distinct methodologies for measuring deforestation, which can result in varying deforestation estimates. PRODES specifically targets clear-cut deforestation of old-growth primary forest, while MapBiomas includes other types of land-cover changes, such as selective logging and secondary forest loss. Furthermore, the spatial and temporal resolutions of the satellite images used by each dataset can impact the detection of deforestation events, particularly when considering small-scale or fragmented forest loss.

These differences in measurement approaches can have important implications for evaluating the effectiveness of deforestation policies in Brazil through three main channels. First, coarser datasets may systematically underestimate total deforestation by failing to detect small patches of forest loss. We refer to this as the measurement issue. This does not necessarily affect policy evaluation in and of itself, but the fact that many patches go undetected can lead to effects on policy evaluation if actors on the ground react strategically to this issue. Second, even if a policy successfully reduces deforestation in larger, more easily detectable patches, datasets with larger minimum detection thresholds may overstate policy effectiveness if deforestation in smaller, undetectable patches persists. This occurs because the data captures only the targeted component of deforestation and misses the residual activity–what we term the loophole effect. Finally, and most pervasively, when deforesters understand the limitations of monitoring technologies, they may deliberately shift their activities toward smaller patch sizes to avoid detection altogether. We call this strategic adaptation.

These three challenges are illustrated in Fig. 1. The first row shows the measurement issue. Panel (a) depicts the actual land use grid, where 16 hectares of forest have been lost across two patches, with a mean patch size of 8 hectares. Panel (b) shows that MapBiomas, with its smaller minimum detection threshold, accurately captures the full extent of deforestation. In contrast, Panel (c) demonstrates that PRODES–due to its higher minimum detection threshold–fails to detect the smaller patch entirely, recording only 12 hectares of loss and inflating the average patch size to 12 hectares. The problem generated here is an underestimation of overall deforestation, which does not per se mean that there will be bias in policy estimation, but it can lead to underestimating how important the environmental issue is to begin with.

The second row of Fig. 1 highlights the loophole effect. Here, deforestation has decreased to just 4 hectares following a hypothetical policy intervention. The land use grid in Panel (d) shows that now only the small patch of deforestation occurs, suggesting

[6] See Appendix Figs. F.1, F.2 and F.3.

[7] For completeness, we report GFW-based estimates in Appendix (F) Table F.1 and Figs. F.2 and F.3. While these results generally align in direction with PRODES and MapBiomas, the effect sizes are smaller and noisier, and GFW lacks patch-level outputs necessary for the spatial and adaptation-focused analyses in Sections 5.2 and 5.3.

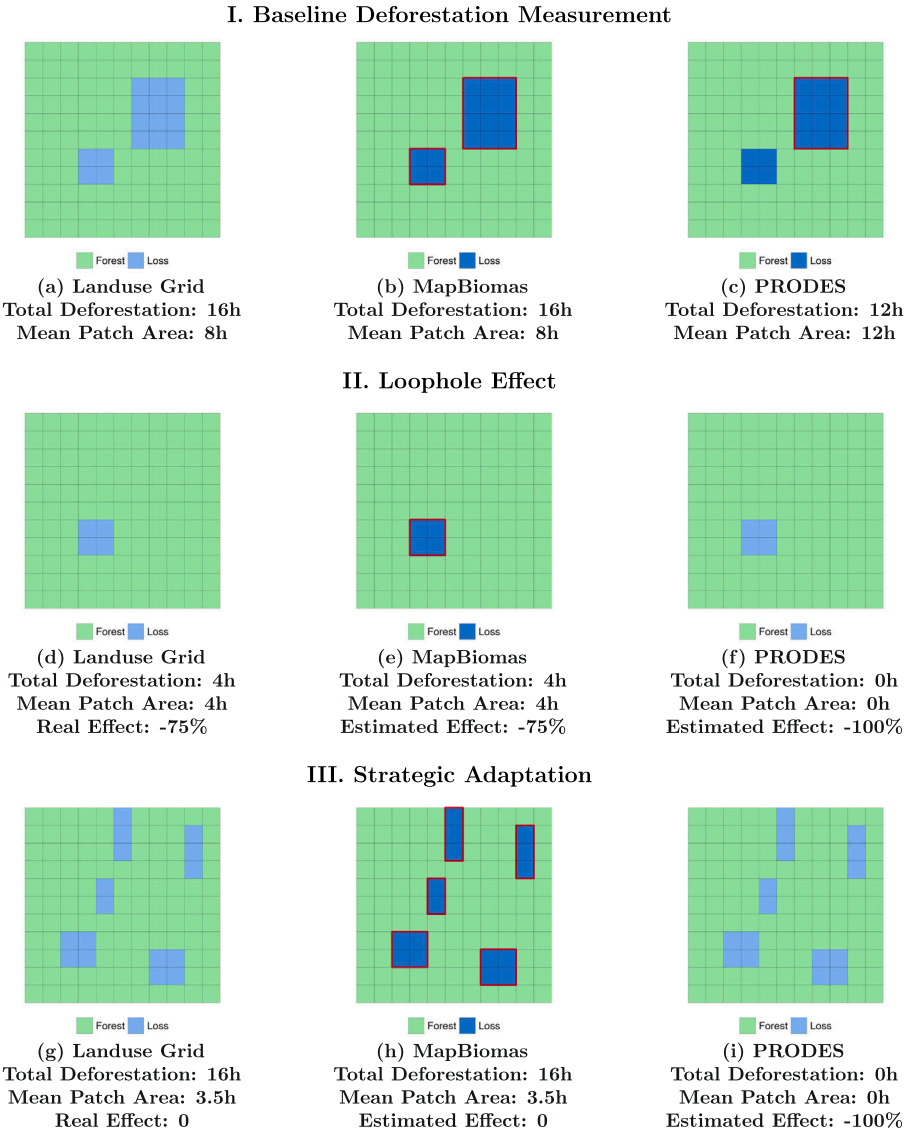


Fig. 1. How Satellite Data's Minimum Detection Threshold Can Affect Outcomes.

that deforesters have responded to the policy by decreasing their large patch deforestation, and not changing their small patch, undetectable, deforestation. Under this scenario, MapBiomas (Panel (e)) detects the full 4 hectares and records a 75 % reduction from baseline. However, PRODES (Panel (f)), which cannot detect patches smaller than 6.25 hectares, records zero deforestation in the post-treatment period and thus estimates a 100 % reduction from pre-treatment deforestation. This creates a misleading picture of policy success, driven not only by changes in actual behavior, but also by the dataset's inability to detect small-scale activity which remains unchanged.

The third row shows an extreme case of strategic adaptation, where agents deliberately fragment their deforestation into small patches, fully substituting large patch deforestation with small patch deforestation. Panels (g)–(i) show the same total deforestation as in the baseline scenario (16 hectares), but broken into numerous smaller patches with a mean size of 3.5 hectares. While MapBiomas continues to detect the full extent of deforestation, PRODES captures none of it. This scenario simulates how actors, aware of PRODES's limitations, might evade detection by strategically adapting their behavior–leading not only to measurement error but potentially to biased estimates of policy effectiveness.

In order for these effects to materialize, deforesting agents must be aware of the limitations of satellite data. The PRODES system's minimum detection threshold of 6.25 hectares is a widely recognized fact within the Brazilian scientific and policy community (Rosa et al., 2012; Richards et al., 2017). As noted by Richards et al. (2017), the use of PRODES for enforcement under programs such as PPCDAm II and the Blacklisting policy created strong incentives for landowners to adapt their behavior in ways that avoid detection. Importantly, this detection threshold became common knowledge among key stakeholders–including cattle ranchers, soy producers,

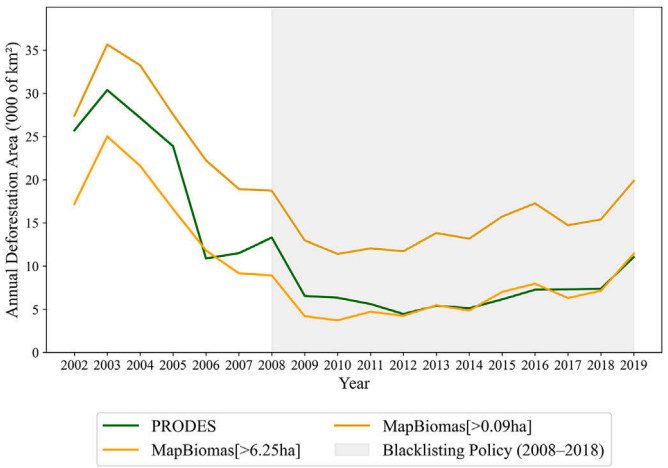


Fig. 2. Yearly deforested area measured by PRODES, and MapBiomas. NOTE: This figure shows total annual deforestation area from 2002 to 2019 as reported by two satellite-based monitoring systems: PRODES and MapBiomas. Both rely on 30-meter Landsat imagery but differ in scope and detection thresholds. PRODES monitors clear-cutting of primary forests and reports patches ≥6.25 ha. In contrast, MapBiomas detects smaller clearings (≥0.09 ha) across all forest types and land categories, including secondary vegetation, without restricting to official forest definitions. The shaded area marks the period of Brazil's blacklisting policy (2008–2018), during which enforcement efforts may have altered the scale and spatial pattern of deforestation.

and smallholder farmers–particularly in regions like northern Mato Grosso and northeastern Pará, where technical expertise and CAR registration were more prevalent. These actors were more likely to be informed about how forest cover was classified and monitored under PRODES. The documented rise in small-patch deforestation (less than or equal to 6.25 ha) in these regions after 2008 further supports this learning and behavioral adaptation. This threshold, along with DETER's operational characteristics, has been extensively discussed in both academic literature and environmental governance discourse (e.g., Rosa et al., 2012; Gollnow et al., 2018 and Instituto Nacional de Pesquisas Espaciais, 2021b). Moreover, the widespread implementation of enforcement mechanisms such as PPCDAm, the Blacklisting Program, the Environmental Rural Registry (CAR), and supply-chain commitments (e.g., Soy Moratorium) has made these detection thresholds common knowledge among key stakeholders–including cattle ranchers, soy producers and smallholder farmers. The implementation of fines on landholders who deforest large swathes of land should further lead to learning among actors on the ground.

Extensive literature shows that the vast majority of deforestation in the region is driven by conversion to cattle pasture, with estimates suggesting that over 70 % of cleared forest is used for low-productivity grazing (Nepstad et al., 2014; Barona et al., 2010). Soy cultivation is the second most significant driver, although it tends to expand into previously cleared pasture rather than directly into forest (Macedo et al., 2012). Our results in Table H.1 and Appendix H confirm that pasture remains the dominant post-deforestation land use in both blacklisted and non-blacklisted municipalities. In this context, it is plausible–and likely–that deforesters are well-informed about the satellite monitoring systems used by enforcement agencies. Previous studies have documented how landholders and illegal clearing operations actively monitor satellite alerts (Assunção and Rocha, 2019; Assunção et al., 2023), while on the ground evidence supports the idea that deforesters are aware and organized (Evaristo, 2023).

2.3. Comparison of the datasets

To measure annual deforestation, we rely on two complementary satellite-based datasets that differ in detection thresholds and classification methods. First, we use Brazil's official estimates from INPE's PRODES program, which monitors clear-cutting of primary forests in the Legal Amazon using Landsat imagery and a minimum detection threshold of 6.25 hectares.[8] Second, we incorporate data from the MapBiomas project, which provides annual land cover and land use change estimates across all of Brazil. Unlike PRODES, MapBiomas captures forest loss of both primary and secondary forest, with a finer detection threshold of 0.09 hectares.[9]

Fig. 2 compares average annual deforestation estimates from PRODES and MapBiomas between 2002 and 2019. Despite differences in measurement scope and sensitivity, both datasets reveal similar temporal patterns: deforestation peaked around 2004, declined steadily through 2012, and then began rising again in the years following the 2012 Forest Code reform. The comparison highlights the robustness of these trends while also underscoring the importance of accounting for differences in detection criteria when interpreting levels of forest loss.

[8] These estimates are available from Brazil's official INPE portal, TerraBrasilis, which provides spatial data from government monitoring programs such as PRODES and DETER. For technical documentation, see Almeida et al. (2013).

[9] We use MapBiomas Collection 8, which was the latest available at the time of analysis. For methodological details, see Souza et al. (2020).

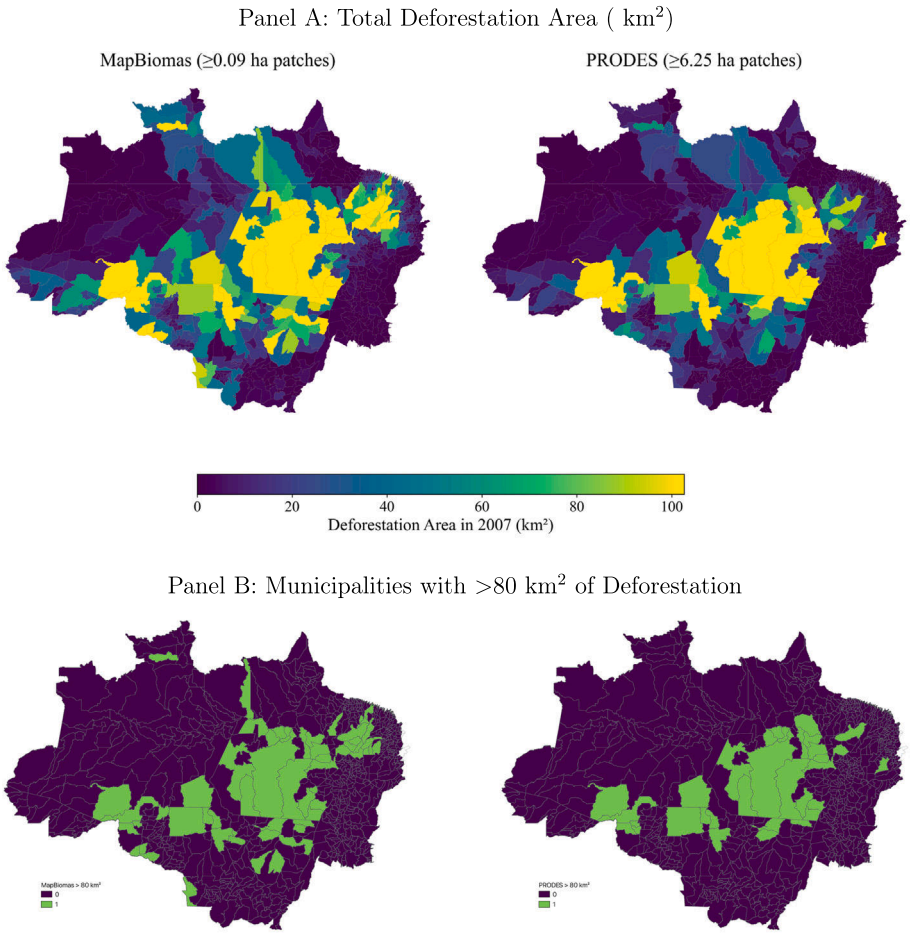


Fig. 3. Municipal-level deforestation in 2007 from MapBiomas and PRODES. NOTE: Panel A compares municipal-level deforestation in 2007 using PRODES (≥6.25 ha patches, left) and MapBiomas (≥0.09 ha patches, right). All maps use a shared color scale capped at the 95th percentile of deforestation to ensure comparability. Cooler tones (blue/purple) indicate low deforestation, while warmer tones (green/yellow) reflect higher levels. Panel B highlights municipalities with over 80 km^2 of deforestation in 2007. MapBiomas identifies 56 such municipalities (26 uniquely), while PRODES identifies 33 (3 uniquely). A total of 30 municipalities exceed the threshold in both datasets, illustrating the additional sensitivity of MapBiomas to small-scale deforestation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In terms of the differences captured by each dataset, we observe that the dataset with lower minimum detection thresholds, Mapbiomas, consistently identifies more hectares of deforestation compared to PRODES, and this difference appears to accentuate after the implementation of the PPCDAm program and the Blacklisting policy in particular. For instance in 2007, the year before blacklisting, PRODES identified a total deforestation of 11,502 km^2, while Mapbiomas captured a total deforestation of 18,926 km^2.

We also replicate PRODES's detection threshold within the MapBiomas dataset by limiting deforestation to patches larger than 6.25 hectares. The resulting series, shown by the dark orange line in Fig. 2, closely tracks the official PRODES estimates. This strong alignment suggests that MapBiomas is not overestimating deforestation through false positives and that the primary difference between the two datasets lies in their sensitivity to small-scale clearing. The comparison highlights the extent of forest loss that goes undetected by PRODES due to its higher minimum mapping threshold, underscoring the value of using data that can capture more granular deforestation dynamics.

Fig. 3 visualizes the spatial distribution of deforestation in 2007, the year prior to the implementation of the blacklisting policy. Panel A shows that while both datasets identify municipalities with high deforestation–particularly in the central region of the deforestation arc–MapBiomas reveals a more extensive pattern of forest loss across the Legal Amazon. In particular, it captures additional clusters of high-deforestation municipalities along the northern and southwestern peripheries, areas that are less visible in the PRODES data due to its higher detection threshold. Panel B further illustrates this difference: MapBiomas identifies 56 municipalities with over 80 km^2 of deforestation in 2007, compared to just 33 in PRODES. Many of these high-deforestation municipalities are only flagged in MapBiomas, highlighting how reliance on PRODES alone may understate the scale and geographic reach of intense deforestation activity.

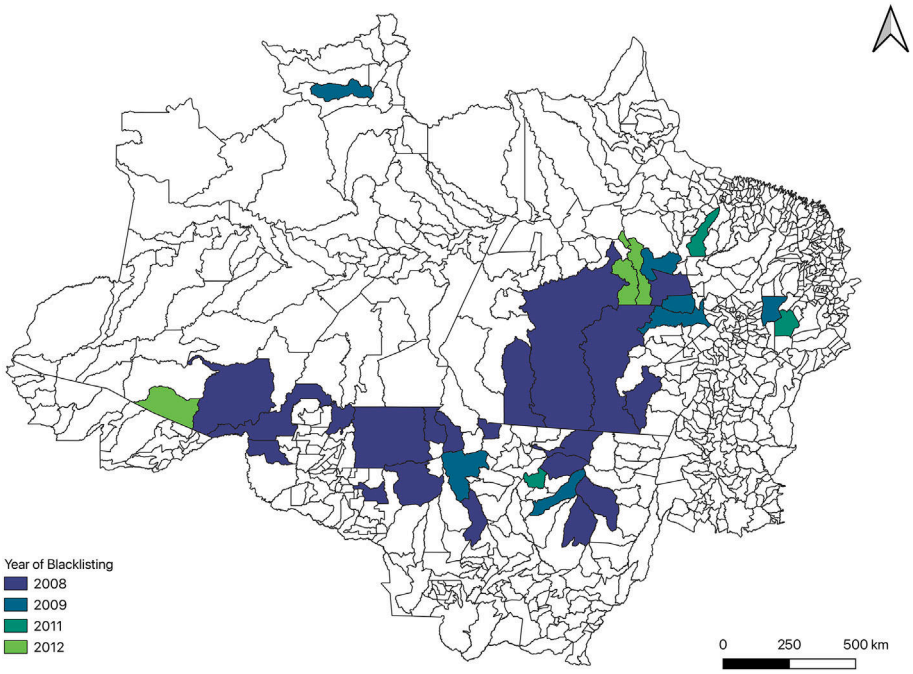


Fig. 4. Blacklisting by Year of Designation. *Note:* This map shows the year of blacklisting for municipalities in the Brazilian Legal Amazon under the federal deforestation monitoring policy. During our study period (2004–2012), 34 municipalities were blacklisted: 21 in 2008, 7 in 2009, 3 in 2011, and 3 in 2012. These designations targeted areas with persistently high deforestation and were intended to restrict rural credit and strengthen enforcement. By 2021, a total of 52 municipalities had been blacklisted, representing 6.7 % of all municipalities in the region.

3. Blacklisting of Brazilian municipalities

The Blacklisting of Municipalities program, also known as the "Priority Municipalities List", is a Brazilian government initiative implemented in 2008 to combat deforestation in the Amazon region (Arima et al., 2014; Cisneros et al., 2015; Assunção and Rocha, 2019; Assunção et al., 2023). The program was established under the federal Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), which aimed to reduce deforestation rates by 80 % by 2020 compared to the 1996–2005 baseline.

Under the Blacklisting program, municipalities with high deforestation rates and environmental crimes were designated as "priority municipalities" and subjected to a series of strict environmental regulations and enforcement actions. The criteria used to determine whether a municipality would be added to the blacklist were established in Decree 6.321/2007: (i) the total area of deforested forest, (ii) the total area deforested in the last three years, and (iii) an increase in the deforestation rate in at least three of the last five years. These criteria were measured using PRODES data, which means that the policy's implementation relied on a dataset that could only detect deforestation patches larger than 6.25 hectares.

The blacklisted municipalities faced a series of consequences, including (i) increased enforcement of environmental regulations, where the Brazilian Institute of Environment and Renewable Natural Resources (IBAMA) intensified monitoring and enforcement actions in these areas, resulting in increased fines and penalties for illegal deforestation activities (Börner et al., 2015); (ii) restrictions on agricultural credit, as the Central Bank of Brazil implemented measures to restrict access to agricultural credit for landowners in blacklisted municipalities who did not comply with environmental regulations (Arima et al., 2014); and (iii) suspension of rural property registration, where the government suspended the registration of new rural properties in these municipalities, hindering landowners from obtaining legal documentation for their land and restricting their access to credit and other resources (Assunção and Rocha, 2019).

Fig. 4 illustrates the staggered rollout of the blacklisting program across municipalities in the Brazilian Amazon. The majority of designations occurred early in the program: 21 municipalities were blacklisted in 2008, followed by 7 in 2009, 3 in 2011, and another 3 in 2012. After 2012, however, the program paused amid growing political resistance and the enactment of the new Forest Code. It did not resume until 2017, when 8 additional municipalities were added, followed by 2 more in 2018. In what follows, we focus our analysis on the period from 2004 to 2012, when the program operated under its original design and enforcement architecture. We expect the bulk of its effects to be concentrated in this initial phase.

Returning to Fig. 3, Panel B, we see that the PRODES data used at the time closely mirrors the actual selection of municipalities into the blacklisting program–most municipalities with over 80 $km^2$of deforestation in 2007 were eventually blacklisted. This alignment

suggests that PRODES effectively guided enforcement decisions. However, MapBiomas data for the same year identify a substantially larger set of high-deforestation municipalities. If MapBiomas had been used as the official monitoring source, it is likely that more municipalities would have qualified for blacklisting. This highlights how measurement thresholds may have constrained the scope of enforcement.

4. Data

To analyze the effects of Blacklisting, we employ synthetic differences-in-differences (Arkhangelsky et al., 2021) in order to account for the fact that municipalities are selected for treatment based on observed past deforestation. Below, we present our data and empirical method.

4.1. Dependent variables

PRODES deforestation rate variable (%) represents the deforestation rate detected by Brazil's official monitoring system, calculated as the ratio between the deforested area identified by PRODES and the total forest coverage in the baseline year 2000, ranging from 0 to 100 percent. This metric specifically captures clear-cut deforestation events larger than 6.25 hectares in primary forest areas, following PRODES' methodology discussed above and recording each deforested patch only once in the year it is first detected. Table 1 shows the values for PRODES total deforestation in km^2, as well as our main dependent variable in percentage points. We can see clearly that treated municipalities have significantly higher values of deforestation in 2007 (the year before most municipalities get treated), with an average deforestation rate of 150 km^2.

MapBiomas deforestation rate variable (%) measures the deforestation rate identified through MapBiomas computed as the percentage of deforested area relative to the total forest extent in 2000, with values between 0 and 100 percent. Unlike PRODES, this variable captures a broader spectrum of forest changes due to MapBiomas' ability to detect smaller-scale deforestation events and its comprehensive land-use classification system that accounts for both primary and secondary forest dynamics at 30-meter resolution. Table 1 shows the values for Mapbiomas total deforestation in km^2, as well as our main dependent variable in percentage points. We can see clearly that treated municipalities have significantly higher values of deforestation in 2007 (the year before most municipalities got treated), with an average deforestation rate of 175 km^2.

4.2. Treatment

We consider a municipality as treated if it is included in the blacklist. The treatment variable is assigned a value of 1 in the year that a municipality enters into the blacklist and remains 1 for subsequent years that the municipality remains blacklisted. For years when a municipality is not blacklisted or removed from the blacklist, the treatment variable is assigned a value of 0. During the study period (2004–2012), 34 municipalities were added to the blacklist: 21 in 2008, 7 in 2009, 3 in 2011, and 3 in 2012. Extending beyond the study period, between 2008 and 2018, a total of 60 municipalities entered the blacklist; of these, 21 eventually exited, while 39 remained blacklisted. These municipalities are shown in Fig. 4, where lighter colors represent earlier blacklisted municipalities while darker ones represent later entrants into the blacklist.

In our analysis, we are focusing on 2004–2012, which contains 34 municipalities that entered the blacklist (considered switchers in the analysis) during this period. Once a municipality enters, we consider it treated for the remaining years in the sample.[10] We decide to focus on this subset of years because 2004 marks a significant change in deforestation policies in Brazil with the implementation of PPCDAm. This makes it more difficult to compare these pre-periods with post treatment periods. We decide to stop our analysis in 2012 because this marks the year that the Forest Code was passed in Brazil, significantly changing the laws around deforestation and land use, and also leading to a relaxation in environmental monitoring, which could have led to the blacklisting program losing efficacy. We provide a set of evidence below showing how deforestation trends change at these important year markers, with 2004, 2008 and 2012 representing important years.

4.3. Covariates

The list of covariates that we include in our models comprises climatic, economic, and policy enforcement variables. Climatic factors are considered a primary driver of forest loss and recovery (Rosa et al., 2024). Earlier research has discussed the local impact of precipitation and temperature variation on deforestation (Anderson et al., 2018; Rosa et al., 2024). Moreover, climatic factors are widely understood as primary drivers to affect secondary forest recovery (Zeng et al., 2019). To account for the effect of precipitation and temperature variation in our analysis, we include municipality-level yearly mean annual precipitation (mm), and temperature (K) constructed using GEE from the TerraClimate database (Abatzoglou et al., 2018).

Population density is a critical factor both for economic outcomes and forest outcomes (Mullan et al., 2018; Gori Maia and Schons, 2020). Population density can significantly influence deforestation pressure, migration patterns, as well as urban growth. We measure population density by dividing number of inhabitants by area of municipality.

Lastly, the area of protected land (under Protected Areas, National Parks or Indigenous Territories) within a given municipality also plays a crucial role in assessing the forest protection policy enforcement. This variable serves a dual purpose: it indicates the

[10] Assunção and Rocha, 2019; Assunção et al., 2023 both use a similar strategy. Note that 2 municipalities were removed from the blacklist by 2012 and a further 4 were removed in 2012.

Table 1
Summary statistics by group.

	Total Sample (N = 6822)		Treated Sample (N = 306)		Untreated Sample (N = 6516)	
Variable	Mean	SD	Mean	SD	Mean	SD
Panel A: 2004–2012 Averages						
Deforested area–PRODES	16.04	53.52	127.91	161.37	9.16	25.04
Deforested area–MapBiomas	24.69	59.71	156.84	163.49	16.57	31.72
Deforested (%)–PRODES	0.62	1.59	1.06	1.09	0.60	1.61
Deforested (%)–MapBiomas	1.65	2.61	1.35	1.14	1.67	2.66
Forested area	4,884.09	13,154.01	20,188.03	28,265.34	4,165.40	11,499.38
Rainfall	5.06	0.27	5.10	0.18	5.06	0.28
Temperature	5.76	0.03	5.77	0.02	5.76	0.03
Protected area	1,563.76	6,402.84	6,920.62	14,632.82	1,312.19	5,611.31
Population density	23.38	126.61	2.82	3.23	24.35	129.47
Population	30,592.88	97,928.46	45,370.02	71,977.76	29,898.92	98,929.94
Panel B: 2007 Cross-Section						
Deforested area–PRODES	15.13	50.23	135.15	152.86	7.76	18.22
Deforested area–MapBiomas	24.90	57.39	156.37	158.30	16.82	29.32
Deforested (%)–PRODES	0.47	0.96	1.05	0.81	0.44	0.96
Deforested (%)–MapBiomas	1.77	3.14	1.30	1.01	1.80	3.20
Forested area	4,884.09	13,161.73	20,188.03	28,643.49	4,165.40	11,506.45
Rainfall	4.97	0.31	5.03	0.21	4.96	0.31
Temperature	5.76	0.03	5.77	0.03	5.76	0.03
Protected area	1,563.38	6,406.68	6,920.62	14,828.59	1,311.80	5,614.84
Population density	22.61	124.54	2.70	3.00	23.54	127.36
Population	29,408.51	94,714.12	43,332.77	67,181.24	28,754.61	95,797.07

NOTE: This table reports municipality-level means and standard deviations (SD) for variables used in the analysis across treated, untreated, and full samples based on 'blacklisting' between 2004 to 2012. MapBiomas values correspond to ≥0.09 ha estimates, while PRODES are by default at ≥6.25 ha. Deforested Area (km^2) indicates annual deforestation, while Forest Area (km^2) is the total forest area at the start of the year. Deforested Share (%) is the annual deforested area as a percentage of the initial forested area. Protected Area (km^2) represents land under legal protection (indigenous lands or conservation units). Rainfall is in millimeters, and temperature in degrees Celsius.

availability of land for protection and also signals the level of monitoring within the municipality, as multiple agencies may be involved in forest protection.

These variables, collected from various sources and databases, capture different dimensions of the socio-economic, environmental, and policy contexts relevant to the analysis. By incorporating these variables into the empirical framework, we aim to better understand the factors influencing deforestation outcomes and evaluate the effectiveness of the policy interventions considered in this study.

Table 1 provides a detailed summary of municipality-level characteristics, including deforestation rates, forested areas, climate variables, and socioeconomic indicators for treated and untreated samples. The treated municipalities, which were added to Blacklist between 2004 and 2012, display significantly higher average deforested areas across both PRODES and MapBiomas sources, with mean values of 143.87 km^2 and 176.95 km^2, respectively. In contrast, untreated municipalities exhibit much lower deforestation levels, averaging 9.95 km^2 (PRODES) and 17.37 km^2 (MapBiomas). Forested areas and protected regions also differ markedly between groups, with treated municipalities possessing larger forested and protected areas on average. These differences across treated and untreated samples are statistically supported, underscoring the distinctive characteristics of municipalities subject to MP policy.

Using this data, our empirical analysis is divided into two main parts. First, in section 4, we revisit the evaluation of the effects of Blacklisting on deforestation outcomes using the two locally calibrated deforestation datasets, PRODES and Mapbiomas. Then, in section 5.2 we employ class level metrics at the municipality by year scale using patch data which we obtain from deforestation rasters of each of the two datasets. We then proceed to test whether blacklisting led to differential patterns of deforestation as measured by these patch derived class-metrics.

4.4. Methods: synthetic difference-in-differences

To evaluate the causal effect of blacklisting on deforestation, we employ the synthetic difference-in-differences (SDID) estimator developed by Arkhangelsky et al., 2021. This approach combines features of both synthetic control and difference-in-differences methods to address potential violations of the parallel trends assumption while allowing for multiple treated units and time periods.

The SDID estimator constructs synthetic control groups by reweighting non-blacklisted municipalities to match the pre-treatment trends of blacklisted municipalities, while simultaneously reweighting pre-treatment time periods to better predict post-treatment outcomes. This double-reweighting approach helps control for both observed and unobserved confounders that may vary across both municipalities and time.

Formally, for municipality i in year t, let Y_{it} denote our outcome measures of deforestation (from MapBiomas and PRODES), let BL_{it} indicate blacklist status, and let X_{it} represent our vector of time-varying covariates including rainfall, temperature, protected areas and population density. The SDID estimator solves:

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau,\mu,\alpha,\beta}{\operatorname{argmin}} \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - \mu - \alpha_i - \beta_t - BL_{it}\tau)^2 \hat{\omega}_i \hat{\lambda}_t \tag{1}$$

where $\hat{\omega}_i$ and $\hat{\lambda}_t$ are municipality and time weights chosen to minimize pre-treatment prediction error while maintaining interpretable scale. The parameter τ captures the average treatment effect of blacklisting on deforestation.

We estimate standard errors using a municipality-clustered bootstrap procedure with 1,000 replications to account for serial correlation in outcomes within municipalities over time. This approach follows Arkhangelsky et al., 2021 and is particularly appropriate given the panel structure of our data.

The key identifying assumption is that, after reweighting both municipalities and time periods, any remaining differences in deforestation trends between blacklisted and non-blacklisted municipalities can be attributed to the causal effect of blacklisting. While this assumption is not directly testable, we provide evidence for its plausibility through extensive pre-trend analyses and placebo tests in the appendix.

This methodology improves upon traditional difference-in-differences by relaxing the parallel trends assumption and upon synthetic control methods by accommodating multiple treated units and time periods while maintaining transparent inference procedures. The approach is particularly well-suited to our setting where treatment timing varies across municipalities and selection into treatment likely depends on past deforestation outcomes. In section D, we report all results using the staggered difference in difference approach proposed by Chaisemartin et al. (2024). All of the signs and magnitudes remain similar. If anything, the staggered difference-in-differences approach shows stronger evidence for strategic adaptation compared to the synthetic difference-in-differences approach, however the staggered difference-in-differences fails to pass the parallel trends test, indicating that synthetic difference in difference is a more appropriate method for our setting.

5. Results

5.1. Do datasets with larger minimum detection thresholds over-estimate the effects of policies?

Table 2, shows the results of our main empirical exercise where we assess the impact of blacklisting on deforestation rates, using different deforestation measures as dependent variables. We employ a synthetic Difference-in-Differences approach (Arkhangelsky et al., 2021). Each column in the table represents estimates based on different sources of deforestation data: PRODES, MapBiomas (MB), and MapBiomas restricted to patches smaller than 6.25 hectares. The dependent variables are measured as a percent deforestation ranging from 0 to 100.

The estimates are consistent in showing a negative impact on deforestation across both full-coverage data sources, implying that blacklisting leads to a significant reduction in deforestation. However, the magnitudes using different sources vary significantly. We find that blacklisting reduced annual deforestation by 0.502 percentage points according to PRODES data, representing a 47.6 % reduction relative to treated municipalities' pre-treatment deforestation rate of 1.08 %. The MapBiomas dataset shows a smaller reduction of 0.398 percentage points (a 31.2 % drop from a 1.30 % baseline). In contrast, when we focus only on small-scale deforestation–MapBiomas patches below 6.25 ha–we observe a statistically significant increase in deforestation by 0.063 percentage points, which corresponds to an 11.1 % increase relative to the baseline mean of 0.566 %. This divergence strongly suggests that blacklisting primarily reduced large-scale detectable deforestation while simultaneously incentivizing a shift toward smaller, less detectable clearing activity. This effect is very close to the 43 % reduction estimated by Assunção et al. (2023), where they use PRODES data and a changes-in-changes approach proposed by Athey and Imbens (2006) to test the effects of blacklisting on deforestation.[11]

Effects measured using MapBiomas data show a reduction of 0.398 percentage points, which corresponds to a 31.2 % reduction relative to their respective pre-treatment baselines of 1.27 % deforestation. These substantial reductions suggest the policy was effective at curbing deforestation, though the magnitude of the effect varies across measurement systems.[12]

The systematic variation in effect sizes across different data sources underscores a critical insight: PRODES overestimates the impact of blacklisting on deforestation compared to datasets with lower minimum detection thresholds. For this systematic overestimation to arise, the deforestation "missed" by PRODES must be correlated with treatment status. Specifically, PRODES must be failing to detect more deforestation in treated municipalities than in untreated ones. This pattern is consistent with both the loophole and the strategic adaptation hypotheses. Under the loophole mechanism, deforesters may continue clearing land below PRODES's detection threshold in areas under stricter enforcement, exploiting the limits of monitoring technology. Under strategic adaptation, this behavior reflects a deliberate response to enforcement incentives: deforesters shift toward smaller, less detectable patches to

[11] Other studies that use PRODES and matching find slightly smaller effects, although they focus on significantly shorter time frames after the policy (2008-2011). Cisneros et al., 2015 find a 24.1 % decrease in deforestation in blacklisted municipalities each subsequent year after treatment, while Assunção and Rocha, 2019 find that the policy prevented 11,218 km^2 of deforestation between 2008 and 2011, representing a 35 % reduction in deforestation.

[12] We also estimate the effects using the GFW dataset and find a non-significant 0.04 percentage point reduction. We decide to drop GFW from the analysis due to the issues with the data which stem from changes in the methodology that make the data to spike during some years (see Figure F-9 in Appendix F for a time series plot including GFW).

Table 2
Effects of Blacklisting on Deforestation (SDID).

Blacklisting (2004–2012)			
	(1) PRODES	(2) MapBiomas	(3) MapBiomas ≤6.25 ha
ATT	−0.502*** (0.107)	−0.398*** (0.099)	0.063** (0.028)
Mean DV Treated Baseline	1.08	1.30	0.566
Effect Relative Baseline	−47.6 %	−31.2 %	+11.1 %
Observations	6,822	6,822	6,822
Switchers	34	34	34
Municipalities	758	758	758
Years	8	8	8
Control variables	Y	Y	Y
Clustered SEs	Munic	Munic	Munic

NOTE: This table presents the results of running synthetic difference-in-differences regressions using Blacklisting (Panel A) as the treatment variable, and three different measures of deforestation: PRODES (≥6.25 ha) (column 1), total MapBiomas (≥0.09 ha) (column 2), and MapBiomas restricted to patches below 6.25 ha (column 3). All regressions include the list of control variables presented in the Covariates section. All regressions include municipality and year fixed effects. Standard errors are in parentheses. * p < 0.05, ** p < 0.01, *** p < 0.001. All dependent variables are log-transformed.

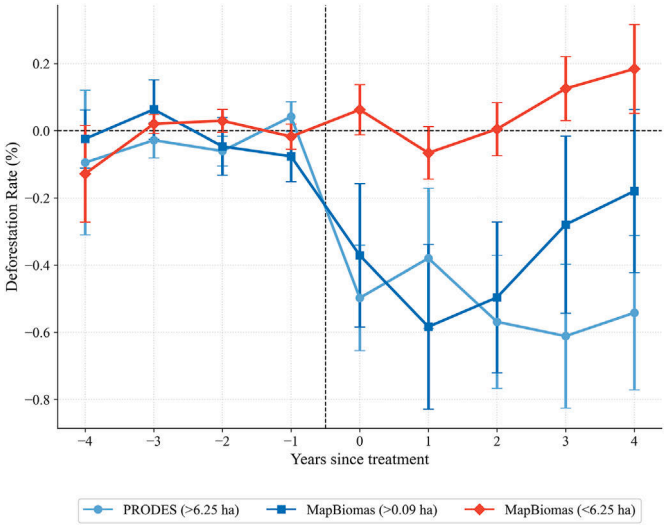


Fig. 5. Effects of Blacklisting on Deforestation Rate. *Notes:* This figure displays the effects of blacklisting on deforestation using synthetic difference-in-differences (SDID). The dependent variable is the deforestation rate as a percentage of forest extent in 2000, as measured by each dataset. The x-axis denotes years since blacklisting. Point estimates and 95 % confidence intervals are plotted. Three deforestation measures are shown: PRODES (≥6.25 ha), MapBiomas (≥0.09 ha), and MapBiomas (≥6.25 ha). The vertical dashed line marks the treatment year (year of blacklisting). Both PRODES and Mapbiomas measures suggest a post-treatment decline in deforestation, though the magnitude and timing vary across datasets and plot size thresholds. Small patch deforestation as measured by MapBiomas (<6.25 ha) shows a slight increase post-treatment, suggesting that most of the difference in the estimates is given by the effects on small patch deforestation.

avoid detection. In both cases, PRODES would overstate the true effectiveness of the policy. In the next section, we explore these mechanisms more directly by examining how the spatial configuration of deforestation changes in response to blacklisting.

5.2. Measuring patterns of deforestation using deforestation patches

In order to identify the differing dynamics in deforestation behavior, we rely on an approach that uses deforestation patches as the main source of information. To do so, we introduce a new method to estimate class-level metrics at the municipality-by-year scale within the Brazilian Legal Amazon using patch data from deforestation rasters. Our approach, depicted in Fig. G.1, builds on the FragStats software package (McGarigal, 1995), which is commonly used in landscape ecology to analyze spatial patterns and landscape structure. We focus on quantifying landscape fragmentation, which occurs when habitats break up into smaller, disconnected patches due to human activities.

We make use of FragStats principles to compute class-level landscape metrics using both the `landscapemetrics` package in R and `PyLandStats`, a Python-based tool that implements a suite of FragStats-like metrics (Bosch, 2019; Hesselbarth et al., 2019). Employing both tools offers complementary advantages: `landscapemetrics` provides a mature, well-documented framework integrated with R's spatial analysis ecosystem, while `PyLandStats` allows for seamless integration into Python-based geospatial pipelines, enabling greater

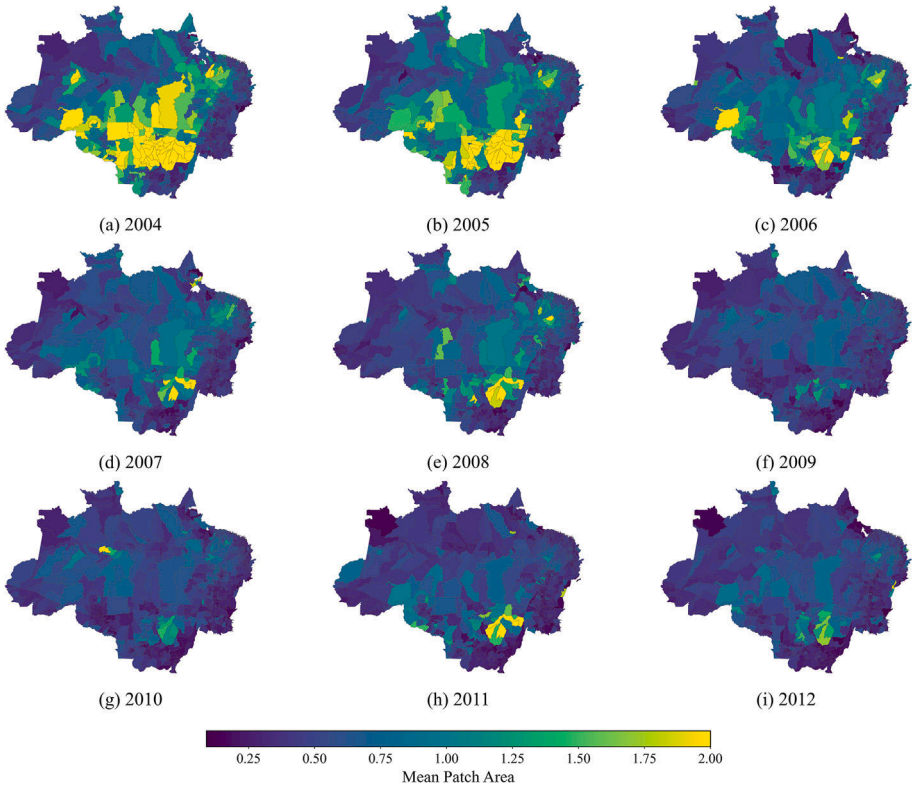


Fig. 6. Mean Patch Area (Mapbiomas) from 2004 to 2012. **NOTE:** This figure shows the Mean Deforestation Patch Area (ha) calculated with Mapbiomas data in each municipality across the years. Lighter (yellow) shades represent larger mean patch areas. You can see a clear decrease in the mean patch area first starting post 2005 with the implementation of PPCDAm and especially markets post 2009 after the implementation of the Blacklisting program. Mean patch area begins to grow again in 2013, in line with weaker environmental enforcement following the 2012 Forest Code and the financial crisis, which meant a disinvestment in IBAMA. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

flexibility in large-scale or automated workflows. Cross-validating metrics across both platforms also enhances the robustness and reproducibility of our landscape analysis.[13]

Specifically, we calculate class-level metrics including the number of patches and mean patch size within each municipality, the proportion of landscape occupied by each land cover class, edge density, fractal dimension, and the landscape shape index. These metrics capture both the composition and spatial configuration of land cover patterns.

In our workflow, depicted in Fig. G.1, we start by leveraging Google Earth Engine to access and recode landuse classification from MapBiomas Collection 8, focusing on distinguishing forested from non-forested areas. This initial step is crucial as it lays the foundation for accurately estimating annual deforestation extents. After recoding, we estimate deforestation, creating binary raster data that highlights deforested versus non-deforested areas based on the different sources of satellite data (PRODES and Mapbiomas). Note that PRODES data has already undertaken the first step of identifying deforestation, so we simply begin with binary rasters produced by these entities.

Once we have our deforestation classifications, we export the binary raster data for further analysis. Shifting to Python, we load the exported data and dive into detailed analysis using tools like PyLandStats. This phase allows us to explore deforestation patterns, understanding not just the where and when of deforestation, but also the geographic distribution of deforestation through various class metrics.

Using these spatial metrics, we can examine whether agents strategically respond to increased enforcement efforts. Given that agents know the government relies on PRODES data for monitoring and enforcement, we expect rational actors to adapt their deforestation practices when monitoring intensifies. Specifically, after policies like municipal blacklisting are implemented, agents may shift from large-scale clearing that PRODES can detect to smaller, fragmented patches that fall below its detection threshold.

In fact, we find preliminary visual evidence that patterns of deforestation in the Brazilian Amazon have shifted over time in response to major policy changes. Fig. 6, which uses MapBiomas data, shows the evolution of mean deforestation patch area at the municipality level from 2004 to 2012. In this figure, darker colors represent larger average patch sizes, while lighter shades indicate

[13] We use popular big-data libraries such as `pandas`, `NumPy`, `Dask`, and `GeoPandas` for data manipulation, processing, and analysis. To handle computational load efficiently, we implement parallel processing using `Joblib`.

smaller patches. We observe a clear decrease in mean patch size starting around 2005, consistent with the rollout of PPCDAm. A second notable decline occurs in 2009–the year following the implementation of the blacklist policy–suggesting that blacklisted municipalities may have adapted by shifting toward smaller, more fragmented clearings.

Complementing this, Fig. E.1 shows that the total number of patches detected by MapBiomas does not decline as sharply, implying that while the total extent of deforestation may have decreased or remained stable, the spatial configuration shifted toward smaller units. In other words, the same or smaller amount of forest was being cleared, but in more fragmented forms. This shift aligns with the interpretation that land users adapted to policy changes by modifying the scale and visibility of deforestation.

Figs. E.2 and E.3, which use PRODES data, offer an important contrast. The patterns of change in patch area and patch counts are less pronounced, especially after 2009. This likely reflects the coarser detection threshold of PRODES, which is less sensitive to smaller-scale clearings. As a result, the finer-grained shifts seen in MapBiomas–particularly the increase in smaller patches–are not as visible in PRODES-based metrics.

Finally, an inflection point appears around 2008, when average patch size begins to rise again in MapBiomas data. This trend coincides with the implementation of the Blacklisting policy, the economic downturn, and weakened enforcement capacity on the ground. Together, these visual patterns suggest that deforestation behavior adapted not only in terms of total area but also in terms of spatial strategy, with smaller, more dispersed patches emerging as a dominant feature during periods of stricter monitoring.

5.3. Evidence from patch level analysis: the loophole effect and strategic adaptation

To investigate how monitoring technology shapes deforestation outcomes, we use blacklisting as an exogenous shift in enforcement intensity and monitoring salience. This exercise allows us to distinguish between two mechanisms: a loophole effect and strategic adaptation. Under the loophole hypothesis, we expect small patch deforestation to continue at similar levels but large-scale deforestation to be reduced by the enhanced monitoring. Under the strategic adaptation hypothesis, we expect regulated actors to actively adjust their behavior in response to enforcement by deliberately fragmenting their clearing activities into smaller, harder-to-detect patches, effectively substituting away from large-scale deforestation and into small-scale deforestation. To evaluate these mechanisms, we begin by estimating the effects of blacklisting on the mean and median deforestation patch size. If either mechanism is at play, we should observe a reduction in average and median patch size following treatment, particularly in municipalities subject to blacklisting (Fig. 7).

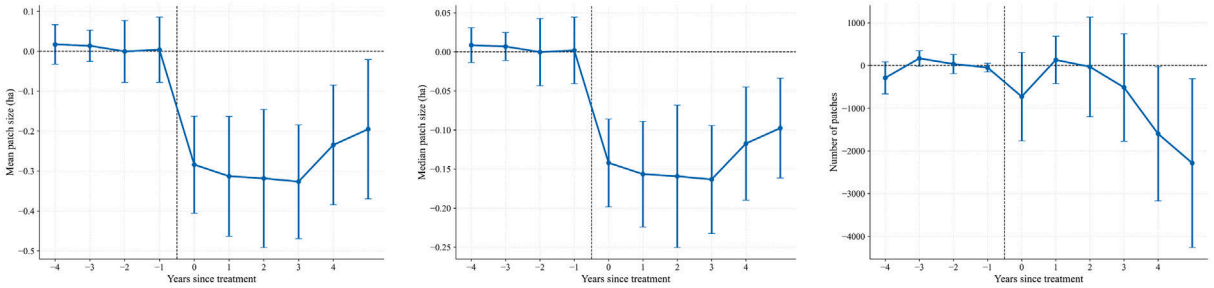
Table 3 shows the effects of blacklisting on the mean patch area, median patch area and the total number of patches. We find that blacklisting significantly reduces deforestation patch sizes. Specifically, the Mean Patch Area shows an ATT of -0.262, translating to a 28.9 % reduction in mean patch area relative to the average patch size in the treated group during the pre-treatment period (baseline mean of 0.907). The Median Patch Area has an ATT of -0.0077, equating to a 7.7 % reduction from the baseline of 0.100, reinforcing the shift toward smaller deforestation patches. The Number of Patches has an ATT of -899.55, indicating a 5.2 % reduction from a baseline of 17,336, though this effect is not statistically significant, suggesting that blacklisting primarily impacts patch size rather than the frequency of deforestation events.

To assess whether blacklisting leads to shifts in the composition and scale of deforestation consistent with either the loophole effect or strategic adaptation, we analyze deforestation outcomes by patch size. Both mechanisms predict a decline in large-scale deforestation following treatment, but differ in their implications for small-scale clearing. Under a pure loophole effect, small-scale deforestation may persist but large patches should decrease. In contrast, strategic adaptation implies that deforesters actively respond to enforcement by fragmenting clearing activity to evade satellite detection while keeping (or only slightly reducing) overall deforestation activity. By examining both the share of total deforestation occurring in each patch size category and the deforestation rate

Table 3
Effects of Blacklisting on Patch Characteristics.

Blacklisting (2004-2012)			
	(1) Mean Patch Area	(2) Median Patch Area	(3) Number of Patches
ATT	−0.262*** (0.0619)	−0.008** (0.0049)	−899.55 (554.89)
Mean DV Treated Baseline	0.907	0.100	17,336
Effect Relative Baseline	−28.88 %	−7.7 %	−5.19 %
Observations	6,822	6,822	6,822
Switchers	34	34	34
Municipalities	758	758	758
Years	9	9	9
Control variables	Y	Y	Y
Clustered SEs	Munic	Munic	Munic

NOTE: This table presents the results of running synthetic Difference-in-differences regressions using Blacklisting (Panel A) as the treatment variable, and Mean Patch Area (column 1), Median Patch Area (column 2), and Number of Patches (column 3) as the outcomes. All regressions include the list of control variables presented in the Covariates section. All regressions include municipality and year fixed effects. Standard errors are in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.



(a) Mean patch size (ha) (b) Median patch size (ha) (c) Number of patches

Fig. 7. Effects of Blacklisting on Patch Characteristics. NOTE: This figure shows the estimated effects of blacklisting on deforestation patch outcomes using a synthetic difference-in-differences (SDID) approach. Each panel reports point estimates with 95 % confidence intervals for years relative to the blacklisting year (indicated by the vertical dashed line). The results show a sharp and persistent decline in both the mean and median size of deforestation patches, along with a delayed but substantial reduction in the number of patches.

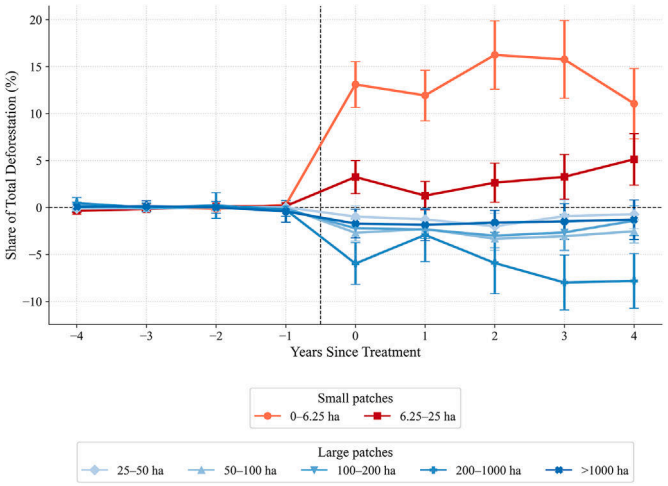


Fig. 8. Effects of Blacklisting on Share of Deforestation From Each Patch Category. *Note:* This figure shows the estimated effects of blacklisting on the relative contribution of different patch size classes to total deforestation area, using a synthetic difference-in-differences (SDID) approach. The vertical dashed line marks the year of blacklisting. Following treatment, the share of deforestation from small patches (especially those <6.25 ha) increases substantially, while the contribution of large patches declines. Table C.2 displays the ATT estimates for each patch size class.

within each size bin, we can distinguish between compositional shifts and behavioral responses. The following figures and tables explore these dynamics, allowing us to evaluate the relative importance of each mechanism.

Fig. 8 presents event study plots from the synthetic difference-in-differences regressions on the share of total deforestation occurring within each patch size bin, allowing us to examine how the composition of deforestation changed following blacklisting. The results provide strong support for the loophole mechanism: the share of deforestation attributable to large patches declines sharply after treatment. Specifically, we observe substantial and statistically significant reductions in the contribution of patches between 50–100 ha (−2.70 percentage points), 200–1000 ha (−5.96 percentage points), and those over 1000 ha (−1.71 percentage points) (Table C.2). At the same time, we find an increase in the share of deforestation from small patches, particularly those below 6.25 ha (+13.10 percentage points) and between 6.25 and 25 ha (+3.25 percentage points). While this compositional shift is suggestive of behavioral adaptation, it is not, on its own, conclusive–these trends may reflect reallocation rather than a net increase in small-scale clearing.

Fig. 9 turns to event study estimates of the deforestation rate itself, by patch size category. These results provide more direct evidence of strategic adaptation. In particular, we observe a statistically significant increase in the deforestation rate for patches smaller than 6.25 ha, with an ATT of 0.063 percentage points—an 11.1 % increase relative to the pre-treatment baseline (Table C.1). Meanwhile, deforestation rates for all large patch categories decline sharply following treatment, with especially pronounced reductions in the 50–100 ha (−0.091 percentage points), 100–200 ha (−0.050), and 200–1000 ha (−0.072) bins. These correspond to relative reductions of 78.4 %, 75.8 %, and 88.6 %, respectively.

Together, the results point to a combination of mechanisms. The sharp and systematic declines in large patch deforestation–visible in both the rate and share of total deforestation–provide compelling evidence that blacklisting successfully deterred large-scale clearing, consistent with a strong loophole effect. At the same time, the increase in small patch deforestation rates, particularly in areas below PRODES's detection threshold, suggests that at least some strategic adaptation is occurring. While much of the observed

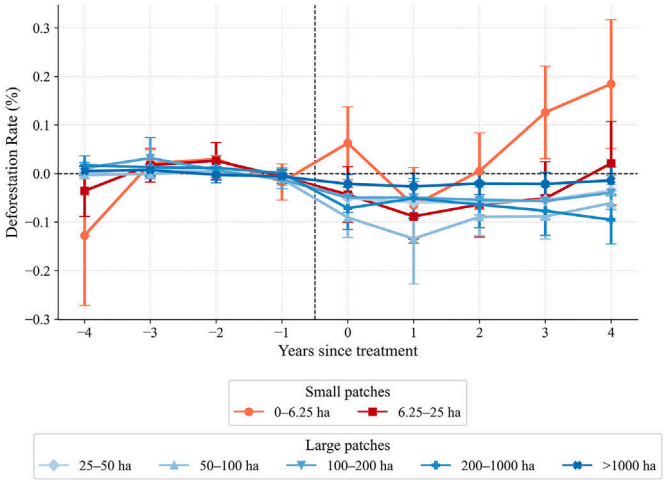


Fig. 9. Effects of Blacklisting on Deforestation Rate by Patch Size. *Note:* This figure shows the estimated effects of blacklisting on deforestation rates by patch size class, using a synthetic difference-in-differences (SDID) approach. The vertical dashed line marks the year of blacklisting. Results indicate a decline in deforestation rates for large patches (≥25 ha) and a relative increase in rates for small patches (<25 ha), especially in the smallest size class (0–6.25 ha). Table C.1 displays the ATT estimates for each patch size class.

policy effect is driven by the deterrence of large-scale deforestation, there is evidence that some deforesters adjusted their behavior by fragmenting clearings to avoid detection thresholds.

6. Conclusion

This paper demonstrates that the choice of satellite data plays a critical role in both the implementation and evaluation of conservation policies. Using the case of Brazil's Blacklisting policy, we show that datasets differ significantly in how they measure deforestation—both in terms of which land cover changes they capture and the minimum patch size required for detection. These differences have substantial implications for how the effects of environmental policies are interpreted.

We first highlight a fundamental measurement issue: satellite-based datasets differ in their ability to detect deforestation, especially at smaller spatial scales. PRODES, the official monitoring system, is restricted to detecting clearings above 6.25 hectares in primary forest. In contrast, MapBiomas identifies deforestation across all forest types and at a much finer scale (down to 0.09 hectares). These differences mean that PRODES systematically underestimates the total amount of deforestation occurring—especially in smaller patches.

Second, we identify what we refer to as a loophole effect. Because PRODES cannot detect forest loss below its minimum detection threshold, deforestation that occurs in smaller patches remains invisible to official systems. Our results show that after blacklisting, deforestation in large patches (especially those greater than 50 hectares) declines significantly, both in absolute terms and as a share of total deforestation. Yet smaller patches continue to be cleared, and much of this activity is missed by PRODES. This loophole effect leads to inflated estimates of policy impact when evaluations rely on coarse detection threshold datasets, even if underlying deforestation dynamics are shifting rather than disappearing.

Finally, we present suggestive evidence of strategic adaptation. Beyond merely exploiting the limitations of PRODES, deforesters may respond to blacklisting by actively shifting their behavior toward smaller-scale clearing to avoid detection and enforcement. We observe an increase in the share and rate of deforestation in the smallest patch category–those under 6.25 hectares–following treatment, consistent with strategic fragmentation. Though these behavioral adjustments do not fully offset the policy's deterrent effect, they pose challenges for enforcement and mislead evaluations based solely on datasets that miss small-scale activity.

Taken together, our findings underscore the importance of using datasets with lower detection thresholds and local calibration–such as MapBiomas–when evaluating deforestation policies. Relying on datasets with coarser detection thresholds not only introduces substantial measurement error but also risks missing meaningful shifts in how deforestation occurs on the ground. For environmental governance to be effective and accountable, particularly in the context of climate change mitigation and biodiversity protection, monitoring systems must capture the full scope of land use change and be robust to the incentives they create for evasion.

CRediT authorship contribution statement

Kathryn Baragwanath: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nilesh Shinde:**

Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude AI and ChatGPT in order to edit the grammar of the writing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Brazil's deforestation monitoring: an overview

Brazil has developed a structure of forest monitoring aiming at various objectives such as monitoring clear-cut, selective logging, and degradation of native vegetation. The central agency responsible for conducting forest monitoring is National Institute for Space Research (Portuguese: Instituto Nacional de Pesquisas Espaciais, INPE) (Almeida et al., 2013). The consolidated program aimed at forest monitoring is the Monitoring Program for the Amazon and Other Biomes (PAMZ +) by the Earth Observation and Geoinformatics Division (DIOTG) and by the Spatial Coordination of the Amazon (COEAM) in the National Institute for Space Research (INPE) (Almeida et al., 2013). It runs three key functional projects for forest monitoring: the Brazilian Amazon Rainforest Monitoring Program by Satellite (PRODES), the Real-Time Deforestation Detection System (DETER), and the land use and land cover mapping system (TerraClass). These three projects collaborate with international agencies like the CBERS1-3 (China-Brazil Earth Resources Satellite), ResourceSat, Landsat, and Sentinel programs.

A.1. INPE's deforestation monitoring system

Table A.1 lists the various government-led programs in forest monitoring. The INPE's flagship program, called PRODES, was launched in 1988 and targets 'clear-cut' deforestation at 20–30 meters resolution of pixels. DETER follows this, targeting daily/monthly 'clear-cut and forest degradation in the Brazilian Amazon'. Additionally, the government has launched various projects monitoring specific aspects of deforestation, ranging from selective logging to forest degradation monitoring. Finally, TerraClass is Brazil's official land use classification project that maintains native and secondary forest vegetation.

PRODES and DETER

INPE's Deforestation Monitoring System–PRODES monitors clear-cut deforestation. It reports annual deforestation rates across Brazilian municipalities since 1988. The data establishes public policy measures such as access to credit, deforestation reduction goals, and funds to invest in deforestation reduction goals. INPE's System supports PRODES for Alerts of Deforestation and Degradation (DETER), launched in 2004. DETER supports actions for controlling deforestation and degradation in Brazilian Amazonia. It gives daily alerts on changes in forest cover for areas larger than 3 hectares (Almeida et al., 2013). The alerts indicate regions that have been entirely deforested (clear cut) and those undergoing forest degradation (logging, mining, burning, and others). An essential contribution of PRODES and DETER could be of interest to Brazil's approach and method of 'forest monitoring to curb deforestation & foster secondary forest growth.' These can be used to measure deforestation controls. REDD+ incentives, greenhouse gas monitoring, and supply chain in agribusiness (Almeida et al., 2013, 2022).

Table A.1
Forest monitoring in Brazil.

Name	Year	Area	Frequency	Key Satellites	Max. Resolution	Key objective
PRODES	1988	Amazon	Annual	Landsat, ResourceSat & CBERS	30	Clear-cut
DETER	2004	Amazon	Monthly	Terra	250	Clear-cut, & Degradation
DEGRAD	2007	Amazon	Annual	Landsat, ResourceSat & CBERS	30	Degradation
TerraClass	2008	Amazon	Bi-annual	Landsat, ResourceSat & CBERS	250	Clear-cut, & Landuse
DETEX	2009	Amazon	NA	Landsat, ResourceSat & CBERS	30	Selective logging
DETER-B	2015	Amazon	Monthly	ResourceSat & CBERS	64	Clear-cut, Degradation, Fire, & Selective logging
GFW	2013	World	Annual	Landsat, Sentinel & Planet	30	Clear-cut
MapBiomas	2015	Brazil	Annual	Landsat, Planet & CBERS	30	Landuse
Brazil Data Cube	2020	Brazil	Annual	Landsat, Sentinel & CBERS	30	Landuse

Notes: The key objective is the author's understanding of the broader mission objective behind the programs. The program aimed at monitoring clear-cut deforestation can support the research on forest degradation, as shown in the exhaustive summary by Almeida et al., 2013 on INPE's three programs MapBiomas (Souza et al., 2020), GFW (Global Forest Watch, 2022), and Brazil data cube by Picoli et al. (2020).

PRODES, since 1988, has provided the official government data on the annual deforestation rates in the Brazilian Legal Amazon (Almeida et al., 2013). The system annually identifies clear-cut deforestation of old-growth forests. Once identified as deforested, such areas are not revisited in the subsequent year mapped, which means that the deforestation rate captured by the PRODES system is the increment of previous years. The mapping year (reference year) comprises August 1 of the prior year to July 31 of the reported year. The annual rate is first estimated using a subset of images from the deforestation hotspots, and the result is published in December each year. Subsequently, the analysis of the complete dataset corresponding to the consolidated data is made available in the first half of the following year. PRODES monitoring system uses 30 m resolution images from Landsat in a combination that seeks to minimize cloud cover interference and ensure the interoperability criteria by using a minimum detection threshold of 6.25 hectares.

TerraAmazon

TerraAmazon is a Geographic Information System (GIS) developed by INPE to support its deforestation mapping projects PRODES, DETER, and TerraClass (landuse mapping) based on aided visual interpretation of images (Almeida et al., 2022). Policymakers and researchers have been using this daily at INPE since 2015 to issue alerts on deforestation and aid the forest monitoring. It includes Brazil's official land use classification maps. Neves et al., 2020 compare the MapBiomas and TerraClass validation, suggesting that TerraClass is a valuable resource for land use classification. However, it is relatively recent and does not provide consistent data access beyond the last decade.

Brazil data cube project

The Brazil Data Cube Project (BDC) is INPE's latest project working with multidimensional data cubes from remote sensing images for entire Brazilian territory (Picoli et al., 2020). The project consists of four main objectives: (1) create Analysis Ready Data (ARD) image collections from medium resolution remote sensing images (10 to 64 m) for the entire Brazilian territory; (2) model these ARD images as multidimensional data cubes with three or more dimensions that include space, time and spectral-derived properties, mainly to support image time series analysis; (3) use, propose and develop big data technologies to create, store, and process these data cubes; and (4) from these data cubes create land use and cover information for Brazil, using time series analysis, machine learning methods, and image processing procedures (Picoli et al., 2020).

A.2. Other independent forest monitoring projects

Mapbiomas

MapBiomas Collections (1-7) to data show the 37 years (1985 to 2021) of Brazil's annual land cover and use maps on a Landsat scale (30 m) with improved accuracy in the biomes and new agriculture classes (Souza et al., 2020). The new limit of biomes on a scale of 1:250,000 from IBGE has already been used. The Mapbiomas is one of the most extensive land use classification projects that estimate forest cover, secondary growth, and primary growth forest cover.

Global forest Watch (GFW)

Hansen Global Forest Change v1.8 shows the 20 years (2000-2020) of tree-cover loss estimates worldwide (Hansen et al., 2013; Global Forest Watch, 2022). The dataset is presented under the Global Forest Watch (GFW) project. It includes the Forest loss during the study period, defined as a stand-replacement disturbance (a change from a forest to a non-forest state). In other words, a user can estimate a year of gross forest cover loss event, i.e., a forest loss during the study period, defined as a stand-replacement disturbance or a change from a forest to a non-forest state and encoded as either 0 (no loss) or a value in the range 1–20, representing loss detected primarily in the years 2001–2020, respectively.

Appendix B. Visual trends

B.1. Evidence of increase of undetectable patches

B.2. Patches: mean patch area (Ha)

Fig. B.1 provides visual support for the main paper's regression findings reported in Table 3. The plots show a clear decline in the mean deforestation patch area in blacklisted municipalities following the 2008 blacklist policy, with a more muted trend in non-blacklisted areas. This visual trend is consistent with the estimated negative and statistically significant ATT on mean patch area reported in column (1) of Table 3 (−0.262, $p < 0.001$), reinforcing the interpretation that blacklisting led to fragmentation of deforestation patches—likely as a strategy to evade detection. The figure complements the causal evidence by illustrating the temporal dynamics and heterogeneity across treatment groups (Fig. B.2).

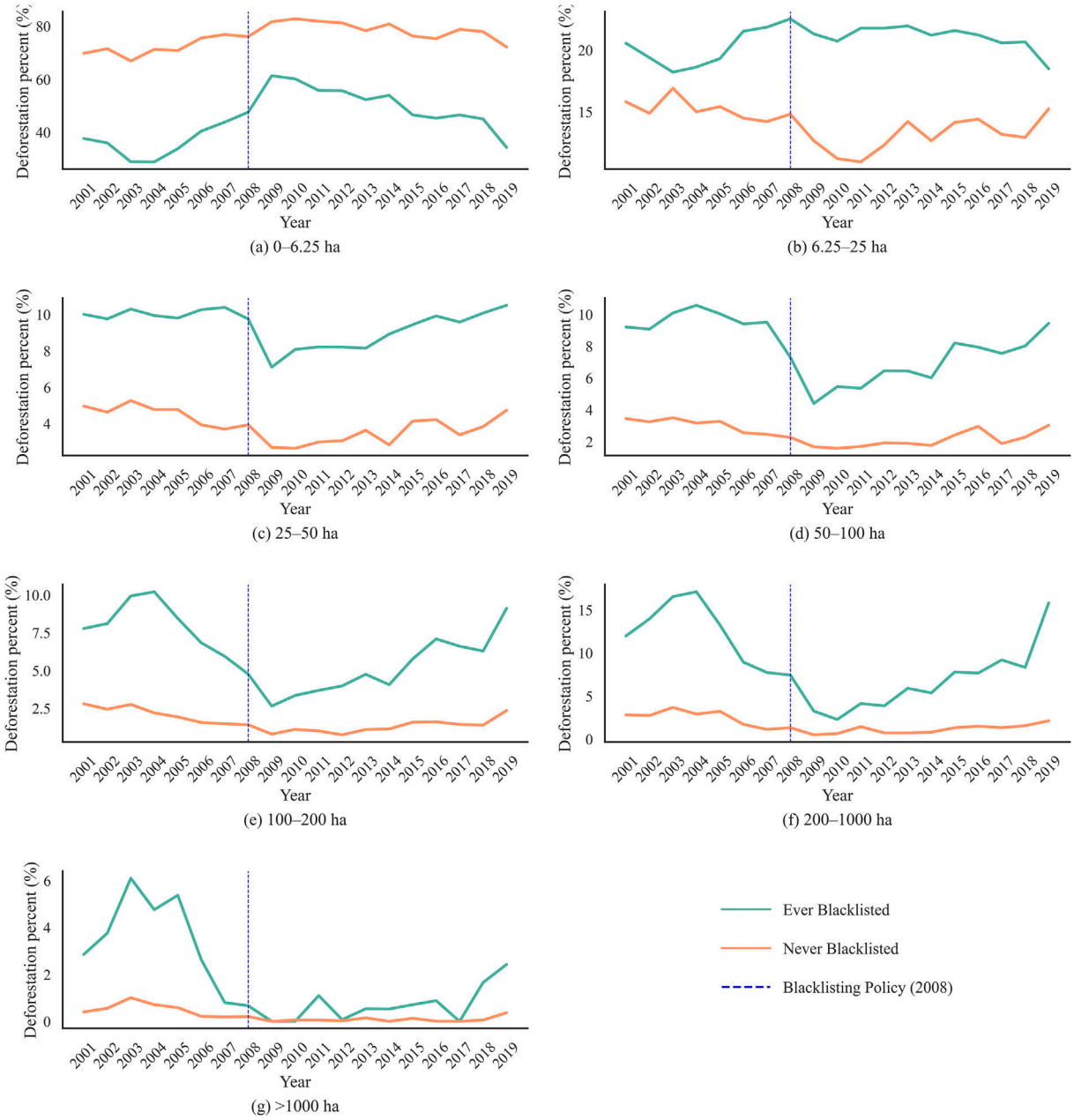


Fig. B.1. Deforestation percent by patch size, Mapbiomas from 2001 to 2019. NOTE: This figure shows the share of total deforestation accounted for by different patch size classes across municipalities, separately for ever-blacklisted and never-blacklisted areas. Each panel reports the percentage of deforestation occurring in a specific patch size category. The vertical dashed line marks the start of the Blacklisting policy in 2008. Following the policy, ever-blacklisted areas show a clear shift in deforestation toward smaller patches, while the distribution in never-blacklisted areas remains relatively stable.

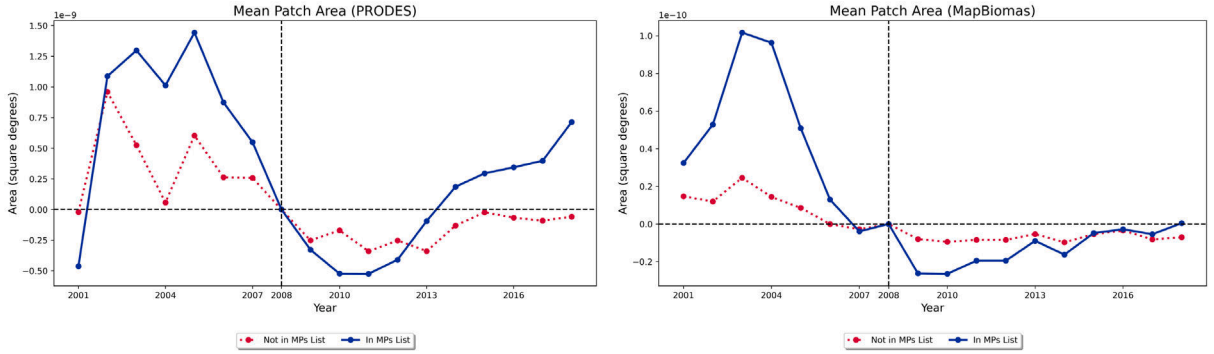


Fig. B.2. Mean Patch Area in Blacklisted vs Non-Blacklisted Municipalities. Note: This figure compares mean deforestation patch areas (in hectares) between blacklisted and non-blacklisted municipalities using data from PRODES (left) and MapBiomas (right). The vertical dashed line indicates the start of the blacklisting policy in 2008. In blacklisted municipalities, the mean patch area sharply declines after 2008, consistent with shifts in behavior to evade detection. In contrast, non-blacklisted municipalities show relatively stable patch sizes over time. The increase in patch size after 2013 corresponds to reduced enforcement following the 2012 Forest Code revision and fiscal cuts to IBAMA.

Appendix C. Tables with estimates

C.1. Deforestation rate by patch size

Interpretation. The blacklisting policy significantly reduced deforestation rates in patches larger than 25 hectares, with particularly large relative reductions for patches over 50 hectares. Effects are strongest in the 50–100 ha and 200–1000 ha bins, suggesting deforesters increasingly avoided large, easily detectable clearings. Small patches under 25 ha saw no significant reduction, and the smallest bin (0–6.25 ha) even shows a positive but insignificant effect—indicative of strategic fragmentation to evade detection.

Table C.1
Effects of Blacklisting on Deforestation Rate by Patch Size (SDID).

Panel A: Patches 0–100 ha				
	(1)	(2)	(3)	(4)
Patch Size (ha)	0–6.25	6.25–25	25–50	50–100
ATT	0.063**	−0.043	−0.051***	−0.091***
	(0.028)	(0.029)	(0.015)	(0.021)
Mean DV Treated	0.566	0.296	0.132	0.116
Effect Relative Baseline	+11.1 %	−14.6 %	−38.5 %	−78.4 %
Observations	6,822	6,822	6,822	6,822
Switchers	34	34	34	34
Municipalities	758	758	758	758
Years	9	9	9	9
Control variables	Y	Y	Y	Y
Clustered SEs	Munic	Munic	Munic	Munic
Panel B: Patches >100 ha				
	(5)	(6)	(7)	
Patch Size (ha)	100–200	200–1000	> 1000	
ATT	−0.050***	−0.072***	−0.022**	
	(0.016)	(0.022)	(0.010)	
Mean DV Treated	0.066	0.081	0.009	
Effect Relative Baseline	−75.8 %	−88.6 %	−248.3 %	
Observations	6,822	6,822	6,822	
Switchers	34	34	34	
Municipalities	758	758	758	
Years	9	9	9	
Control variables	Y	Y	Y	
Clustered SEs	Munic	Munic	Munic	

NOTE: This table shows the estimated treatment effects of blacklisting on the deforestation rate, disaggregated by patch size using synthetic Difference-in-Differences (SDID). Standard errors are in parentheses. * p < 0.05, ** p < 0.01, *** p < 0.001.

C.2. Share of deforestation by patch size

Interpretation. Blacklisting caused a notable redistribution in the spatial pattern of deforestation. The share of deforestation in patches smaller than 25 ha increased significantly, particularly in the 0–6.25 ha bin (+30 %), while the share in all larger patch size bins declined, with the steepest drops observed for patches >200 ha (−77 %) and >1000 ha (−215 %). This provides strong evidence of strategic fragmentation, as deforesters increasingly concentrated activity into small, less detectable clearings.

Table C.2
Effects of Blacklisting on Share of Total Deforestation by Patch Size (SDID).

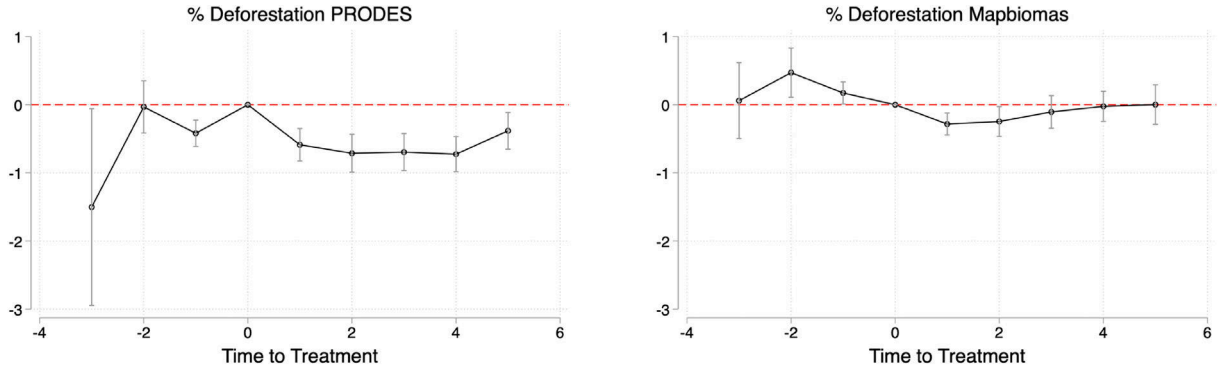
Panel A: Patches 0–100 ha				
	(1)	(2)	(3)	(4)
Patch Size (ha)	0–6.25	6.25–25	25–50	50–100
ATT	13.10***	3.25***	−0.97	−2.70***
	(1.25)	(0.90)	(0.59)	(0.44)
Mean DV Treated	43.81	21.85	10.38	9.50
Effect Relative Baseline	+29.9 %	+14.9 %	−9.4 %	−28.4 %
Observations	6,822	6,822	6,822	6,822
Switchers	34	34	34	34
Municipalities	758	758	758	758
Years	9	9	9	9
Control variables	Y	Y	Y	Y
Clustered SEs	Munic	Munic	Munic	Munic
Panel B: Patches >100 ha				
	(5)	(6)	(7)	
Patch Size (ha)	100–200	200–1000	>1000	
ATT	−2.22***	−5.96***	−1.71**	
	(0.52)	(1.13)	(0.78)	
Mean DV Treated	5.93	7.73	0.80	
Effect Relative Baseline	−37.4 %	−77.1 %	−214.5 %	
Observations	6,822	6,822	6,822	
Switchers	34	34	34	
Municipalities	758	758	758	
Years	9	9	9	
Control variables	Y	Y	Y	
Clustered SEs	Munic	Munic	Munic	

NOTE: This table shows the estimated effects of blacklisting on the share of total deforestation that occurs in each patch size category using SDID. Standard errors are in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix D. DiD using de Chaisemartin and D'Haultfoeuille (2024)

D.1. PRODES vs mapbiomas

See Fig. D.1.

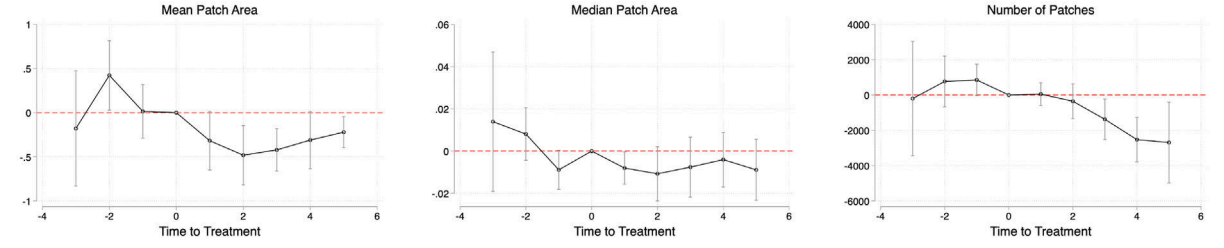


(a) PRODES Deforestation (b) Mapbiomas Deforestation

Fig. D.1. Difference-in-Differences Analysis of Deforestation Rates.

D.2. Patch analysis: mean, median, N

See Fig. D.2.

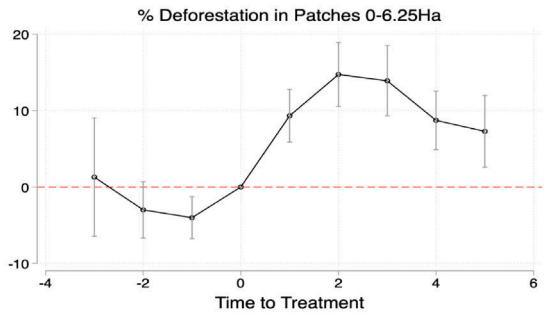


(a) Mean Patch Area (b) Median Patch Area (c) Number of Patches

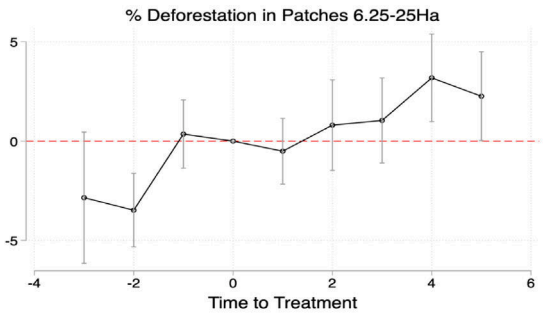
Fig. D.2. Analysis of Forest Patch Characteristics.

D.3. Patch sizes

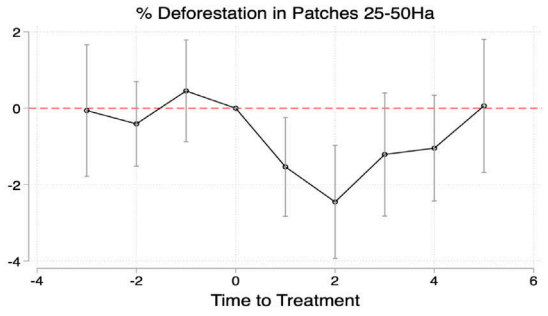
See Fig. D.3.



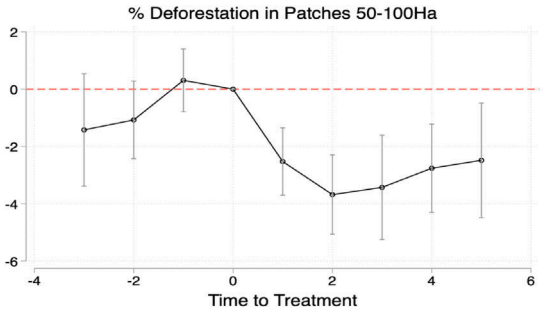
(a) 0-6.25 Ha



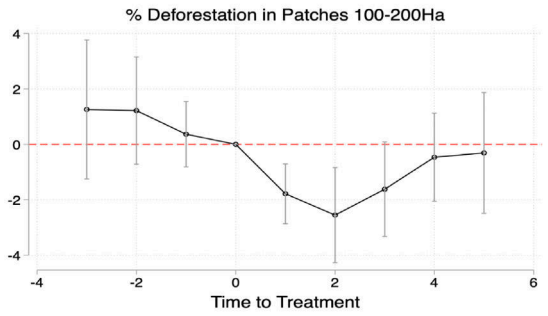
(b) 6.25-25 Ha



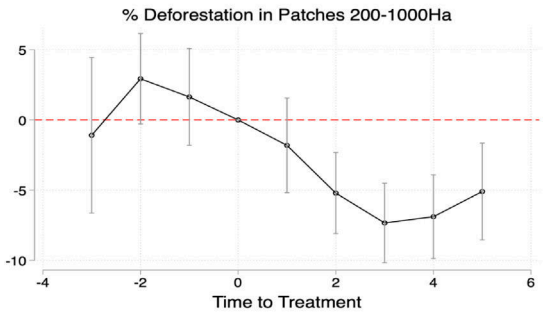
(c) 25-50 Ha



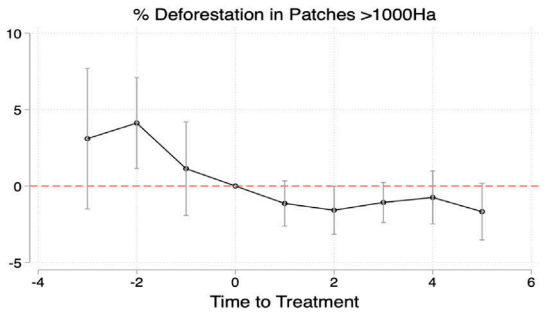
(d) 50-100 Ha



(e) 100-200 Ha



(f) 200-1000 Ha



(g) >1000 Ha

Fig. D.3. Deforestation Patterns by Patch Size.

Appendix E. Patch metrics maps

E.1. Using MapBiomas

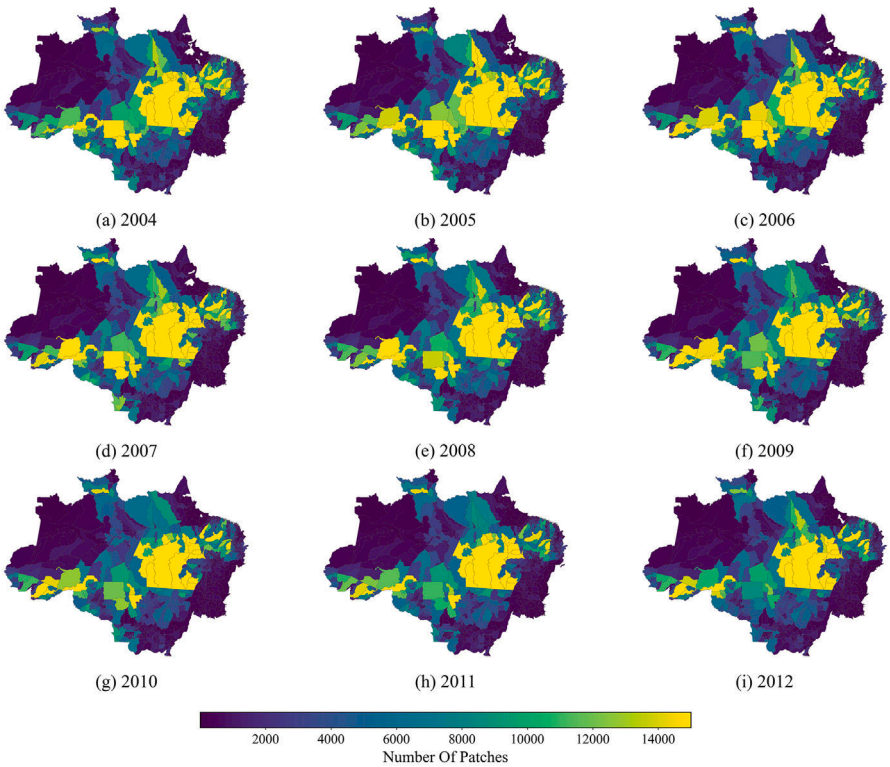


Fig. E.1. Number of Deforestation Patches (MapBiomas), 2004–2012. NOTE: This figure shows the total number of deforestation patches per municipality from 2004 to 2012 using MapBiomas data. Each panel corresponds to a different year, with color intensity reflecting the number of detected patches–brighter yellow areas indicate more fragmented deforestation activity. Compared to PRODES, MapBiomas captures a greater number of small-scale clearings due to its lower detection threshold. The figures show a reduction in patch counts after 2005, likely influenced by early enforcement actions such as PPCDAm. A sharp decline in 2009 aligns with the introduction of the Blacklist policy, suggesting shifts in deforestation behavior. However, fragmentation persists across municipalities, and the spatial patterns indicate that small-scale deforestation remained widespread even under formal enforcement regimes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

E.2. Using PRODES

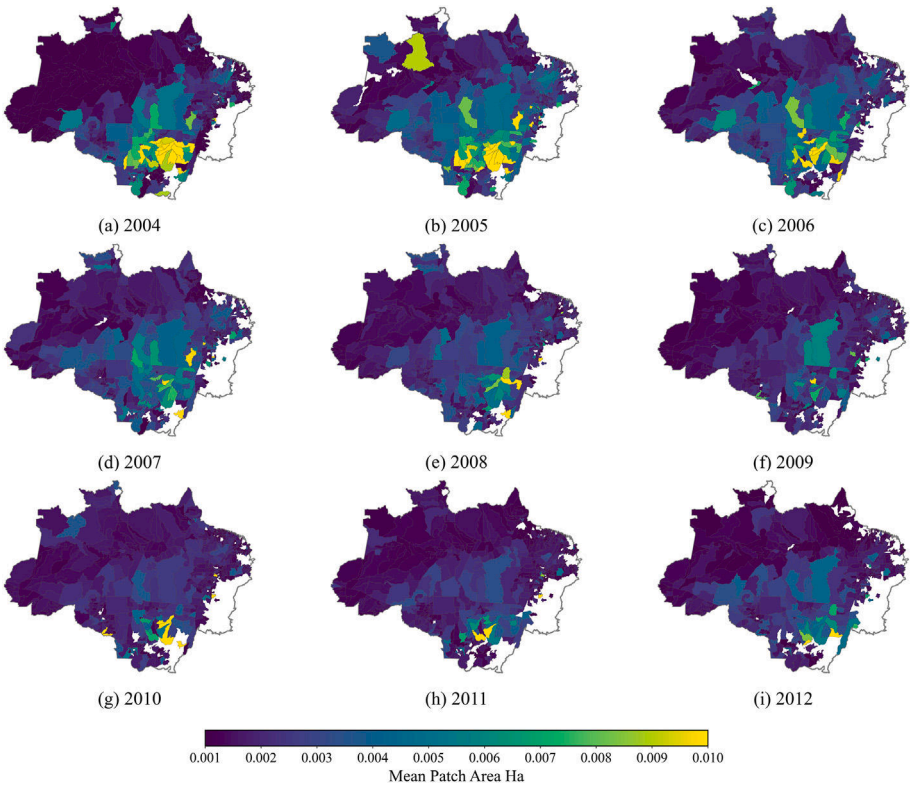


Fig. E.2. Mean Patch Area (PRODES), 2004–2012. NOTE: This figure presents the mean deforestation patch area (in hectares) per municipality from 2004 to 2012 using PRODES data. Each panel corresponds to a specific year, with darker shades representing larger average patch sizes. The maps reveal a clear decline in patch size following 2005, consistent with the implementation of stricter monitoring and enforcement through PPCDAm. A further reduction is evident after 2009, aligning with the rollout of the Blacklist policy, which likely incentivized smaller-scale, less detectable clearings. After 2012, however, patch sizes begin to rise again–particularly in some southern and eastern Amazon regions–suggesting a loosening of enforcement pressure, possibly linked to the 2012 Forest Code revision and budgetary constraints facing IBAMA.

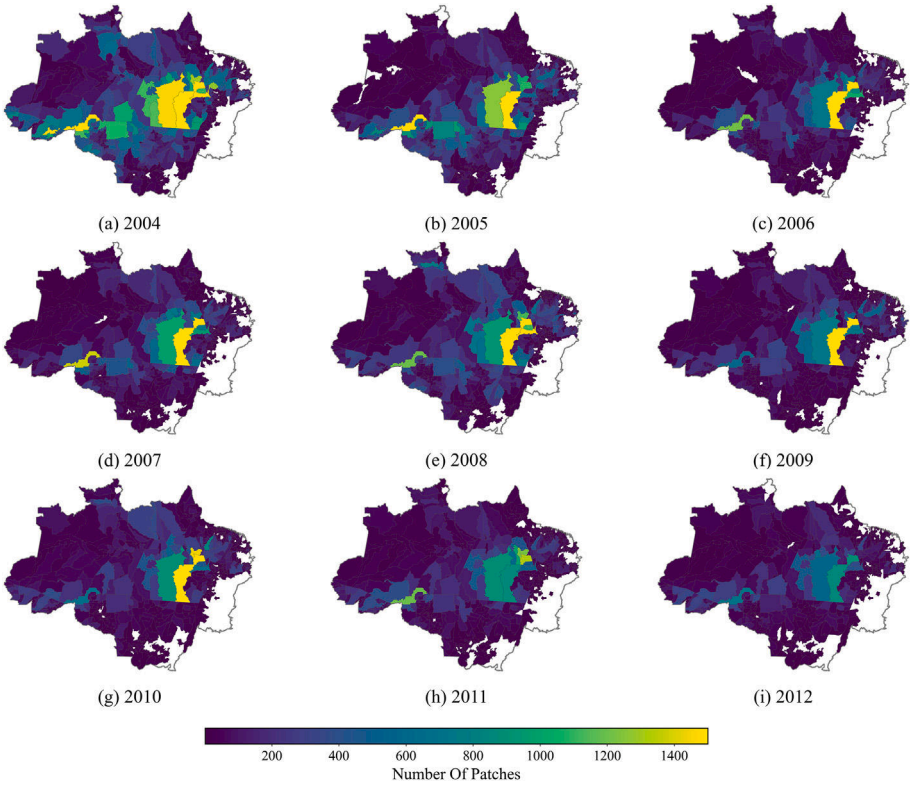


Fig. E.3. Number of Deforestation Patches (PRODES), 2004–2012. NOTE: This figure displays the total number of deforestation patches per municipality across selected years from 2004 to 2012, based on PRODES satellite data. Each panel represents a different year, with color intensity indicating the number of patches–brighter yellow areas denote more numerous and fragmented deforestation events. A notable decline in patch counts begins after 2005, likely reflecting the effects of enforcement efforts such as the PPCDAm plan. The sharp drop observed in 2009 coincides with the implementation of the Blacklist policy, suggesting a reduction in detectable deforestation activity or shifts in land-clearing behavior. In contrast, the gradual increase in fragmentation visible from 2010–2012 may reflect waning enforcement capacity, potentially due to institutional weakening after the 2012 Forest Code revision and federal budget constraints. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Appendix F. Global forest watch dataset

While Global; hence we have renumbered all figures. Please check. Forest Watch (GFW) provides a widely used global measure of tree cover loss, we do not include it in our main analysis due to substantial temporal inconsistencies and methodological limitations. As shown in Fig. F.1, the deforestation rate and total area reported by GFW exhibit erratic year-to-year variation, particularly after 2011. These fluctuations are not mirrored in either PRODES or MapBiomas, and have been noted in prior studies (e.g., Turubanova et al. (2018); Curtis et al. (2018)). A key contributor to this instability is a series of methodological revisions to GFW's loss classification algorithms around 2011 and again in subsequent years (Global Forest Watch, 2017), which introduced breaks in comparability over time. In addition, GFW includes losses from plantation forests, secondary vegetation, and fire-related disturbances—many of which do not correspond to Brazil's legal definition of deforestation.

These discrepancies are further evident in both event-study and average treatment effect estimates. Fig. F.3 shows that PRODES and MapBiomas detect a sharp and sustained decline in deforestation rates following blacklisting, while the GFW-based estimates appear highly variable and fail to capture a comparable post-treatment trend. Table F.1 confirms this: while the synthetic difference-in-differences (SDID) estimates for PRODES and MapBiomas are large and statistically significant, the estimated effect using GFW is smaller in magnitude and statistically indistinguishable from zero.

Taken together, these findings reinforce our decision to rely on PRODES and MapBiomas for the main empirical analysis. GFW's broader detection scope, algorithmic instability, and divergence from Brazil's policy-relevant forest definitions make it less suitable for evaluating the enforcement effects of the blacklist policy. Nonetheless, for transparency and completeness, we report parallel GFW-based estimates in Appendix Table F.1 and Fig. 5.

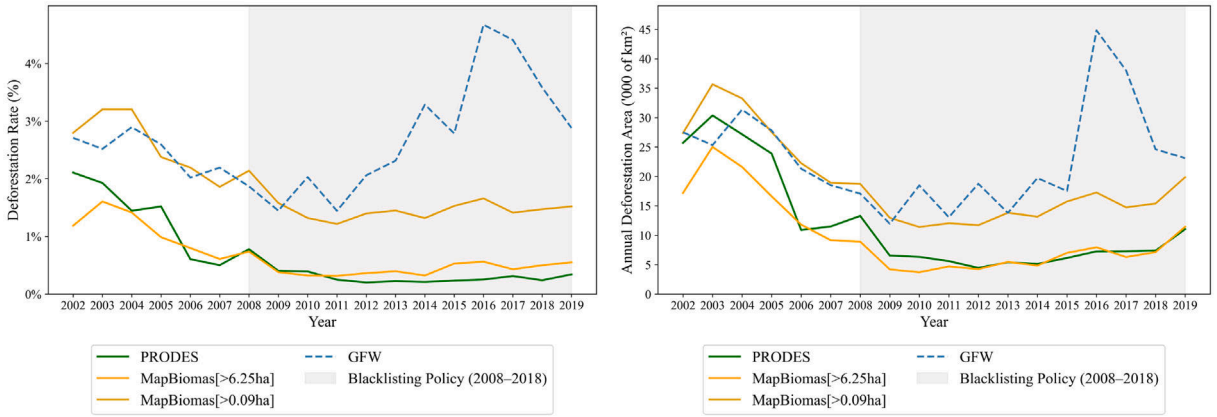


Fig. F.1. Yearly Deforested Area Measured by PRODES, MapBiomas, and GFW. NOTE: This figure presents trends in total deforestation area and deforestation rate across Brazilian Amazon municipalities from 2002 to 2019, based on three satellite-based monitoring systems: PRODES, MapBiomas, and Global Forest Watch (GFW). All three rely on 30-meter Landsat imagery but differ in scope and detection thresholds. PRODES reports only clear-cutting of primary forests ≥6.25 hectares. MapBiomas detects smaller-scale clearings (≥0.09 ha) across all vegetation types and land use classes, including secondary vegetation, without limiting to official forest definitions. GFW data also includes forest loss from plantation and non-primary forests and adopts a globally consistent classification. The top panel shows the deforestation rate as a percentage of forest area, while the bottom panel shows the annual deforested area in thousands of km^2. The shaded area indicates the period of the Blacklisting policy (2008–2018), when targeted enforcement actions were in place. All sources show a decline after 2004–2005, sharper in PRODES, followed by divergence in later years–particularly due to changes in GFW's algorithm post-2011 and broader detection scope in MapBiomas.deforestation.

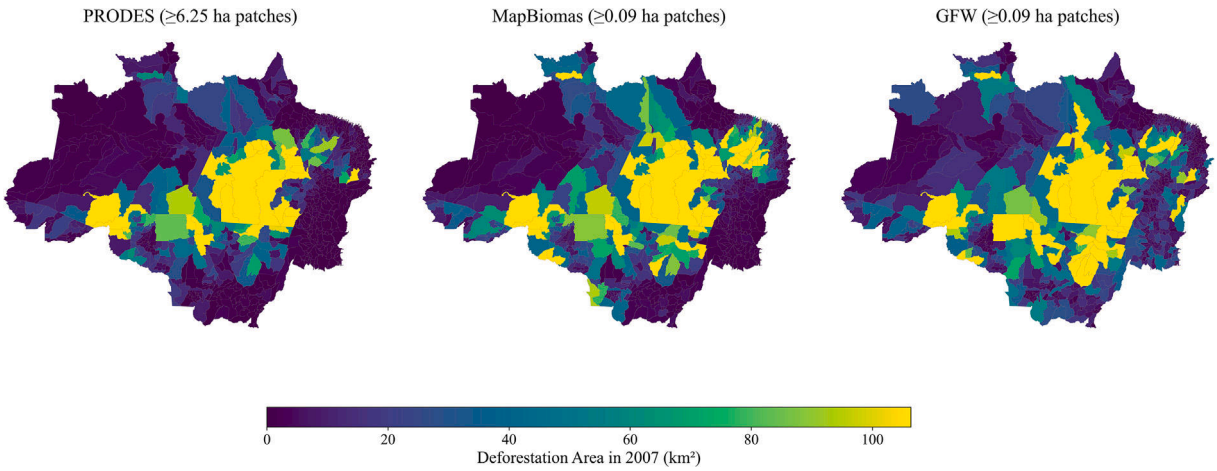


Fig. F.2. Deforestation in 2007 Across Three Datsets. NOTES: This figure compares the spatial distribution of deforestation in 2007 using three satellite-based monitoring systems: PRODES (≥6.25 ha patches), MapBiomas (≥0.09 ha patches), and Global Forest Watch (GFW, ≥0.09 ha patches). Each panel maps the total deforested area per municipality using a consistent color scale capped at the 95th percentile to enhance comparability. While all sources identify core hotspots of forest loss in central and southeastern Amazon regions, PRODES and MapBiomas differ in patch size thresholds and land cover classification, and GFW includes additional sources of tree loss, such as understory fire. The spatial overlap is high across systems in high-deforestation areas, but notable differences emerge in regions with small-scale clearings or non-primary vegetation, highlighting methodological differences in detection scope and forest definitions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table F.1
Effects of Blacklisting on Deforestation (SDID).

Blacklisting (2004–2012)			
	(1) PRODES	(2) MapBiomas	(3) GFW
ATT	−0.502*** (0.107)	−0.398*** (0.099)	−0.082 (0.138)
Mean DV Treated Baseline	1.08	1.30	2.64
Effect Relative Baseline	−47.6 %	−31.2 %	−6.9 %
Observations	6,822	6,822	6,822
Switchers	34	34	34
Municipalities	758	758	758
Years	8	8	8
Control variables	Y	Y	Y
Clustered SEs	Munic	Munic	Munic

Note: This table presents the results of running synthetic difference-in-differences regressions using Blacklisting (Panel A) as the treatment variable, and PRODES (column 1), MapBiomas (column 2), and GFW (column 3) as alternative deforestation measures. All regressions include the list of control variables presented in the Covariates section. All regressions include municipality and year fixed effects. Standard errors are in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All dependent variables are log-transformed.

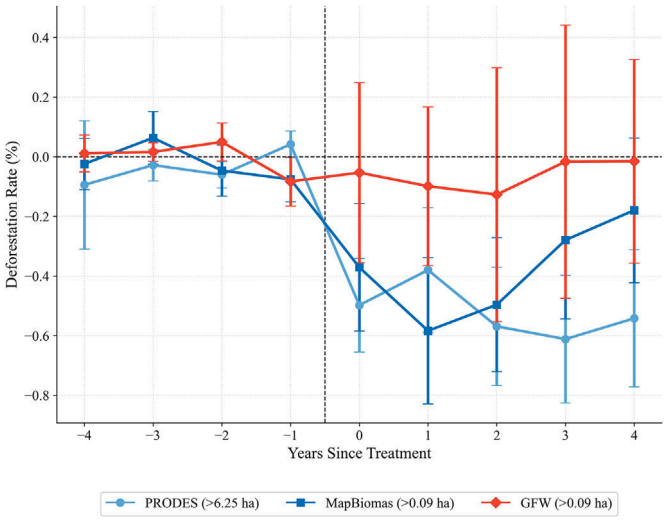


Fig. F.3. Effects of Blacklisting on Deforestation Rate. Notes: This figure shows the event study estimates of the impact of blacklisting on deforestation rates using three satellite-based datasets: PRODES, MapBiomas, and Global Forest Watch (GFW). The x-axis represents years relative to blacklisting implementation, while the y-axis shows changes in the deforestation rate (in percentage points). The plotted lines represent average effects, and the vertical bars indicate 95 % confidence intervals. Both PRODES and MapBiomas detect a sharp decline in deforestation rates beginning in the year of treatment (year 0), with sustained reductions over the following years, suggesting that blacklisting had a significant deterrent effect. In contrast, the GFW-based estimates are noisier and do not show a comparable post-treatment decline, likely due to methodological differences and greater sensitivity to small-scale or non-primary forest loss. These discrepancies reinforce the importance of carefully choosing satellite products aligned with Brazil's legal and enforcement definitions when evaluating deforestation policy impacts.

Appendix G. Notes on deforestation patch estimation

Our workflow is designed to estimate both patch-level and landscape-level metrics for all municipalities within the Legal Amazon region. The primary goal is to compute these metrics consistently across multiple years using a structured, scalable, and fully reproducible process.

The workflow consists of two main stages: (1) Data Preparation and (2) Analysis.

Stage 1: Data Preparation. We begin by using Google Earth Engine (GEE) to process annual land-use and land-cover classification maps. These maps are harmonized and preprocessed to focus on forest dynamics. Specifically, all land cover classes corresponding to forests are recoded to 1, while all non-forest classes are recoded to 0. We then compute annual deforestation by comparing binary forest maps from consecutive years. A pixel is labeled as "deforested" (value $= 1$) if it transitions from forest (1) in year t to non-forest (0) in year $t + 1$. All other pixels are labeled as 0. The resulting binary raster, where 1 indicates deforestation and 0 otherwise, is exported for further analysis.

Stage 2: Metric Computation and Analysis. The binary deforestation rasters are loaded into Python and R for analysis. We make use of FragStats principles to compute class-level landscape metrics using both the `landscapemetrics` package in R and `PyLandStats`, a Python-based tool that implements a suite of FragStats-like metrics (Bosch, 2019; Hesselbarth et al., 2019). Employing both tools offers complementary advantages: `landscapemetrics` provides a mature, well-documented framework integrated with R's spatial analysis ecosystem, while `PyLandStats` allows for seamless integration into Python-based geospatial pipelines, enabling greater flexibility in large-scale or automated workflows. Cross-validating metrics across both platforms also enhances the robustness and reproducibility of our landscape analysis.[14]

We compute a comprehensive set of landscape metrics:

- **Patch-level metrics:** Number of patches, mean and median patch size within each municipality;
- **Class-level configuration metrics:** Proportion of landscape occupied by each land cover class, edge density, fractal dimension, and the landscape shape index;
- **Landscape complexity metrics:** Marginal entropy, conditional entropy, joint entropy, and mutual information, as described in Nowosad and Stepinski (2019).

All computed metrics are compiled into structured CSV files, which serve as the basis for further statistical analysis and policy evaluation.

Reproducibility and Automation. This workflow integrates the strengths of Google Earth Engine (for large-scale geospatial preprocessing), Python (for data manipulation and automation), and R (for landscape metrics computation) within a version-controlled environment. This ensures that the analysis is not only scalable to long time series and large geographic extents, but also fully transparent and reproducible.

Data Management Raw and processed data are organized as follows:

- `data/raw`: Original land-use rasters (e.g., MapBiomas, PRODES), named `{source}_{year}.tif`.
- `data/processed`: Binary deforestation masks with forested pixels $= 0$ and deforested $= 1$.
- `results/metrics`: Final CSV tables of all computed metrics.
- `metadata/deforestation_metadata.csv`: Log of data sources, CRS (EPSG:4326), resolution, and processing timestamps.

Data Preparation

1. **Import and Recode (GEE):** Load annual land-use layers into Google Earth Engine. Recode forest classes to 1, and others to 0.
2. **Detect Deforestation:** Compare the binary maps of year t and $t - 1$. Flag pixels that change from 1 (forest) to 0 (non-forest).
3. **Export Masks:** Save each yearly deforestation raster as `deforest_{source}_{year}.tif` to Google Cloud Storage, and then download to `data/processed`.

Spatial Analysis

We use Python (`rasterio`, `numpy`, parallelized with `dask` or `joblib`) to:

- Clip each binary mask by municipality boundaries from IBGE shapefiles.
- Compute metrics using the `landscapemetrics` R package Hesselbarth et al., 2019 (drop-in for FRAGSTATS (McGarigal, 1995)):
  - Total deforested area per municipality-year.
  - Number of deforestation patches.
  - Mean and median patch sizes.

[14] We use popular big-data libraries such as `pandas`, `NumPy`, `Dask`, and `GeoPandas` for data manipulation, processing, and analysis. To handle computational load efficiently, we implement parallel processing using `Joblib`.

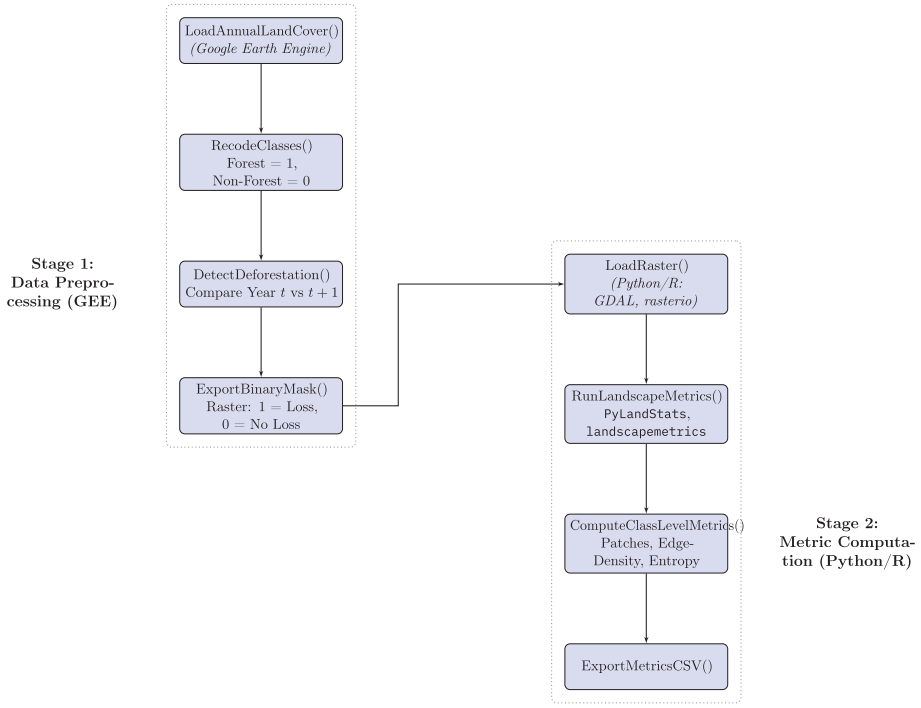


Fig. G.1. System workflow for landscape-level deforestation analysis. The pipeline consists of two main stages: (1) data preprocessing in Google Earth Engine to generate annual binary forest loss rasters; and (2) landscape metric computation using `PyLandStats` (Python) and `landscapemetrics` (R). Outputs are structured CSVs suitable for downstream statistical analysis.

  - Landscape complexity: marginal entropy, conditional entropy, joint entropy, and mutual information Nowosad and Stepinski, 2019.

- Organize results into a tidy `pandas.DataFrame` (one row per municipality-year).

Export of Results

The final metrics are written to:

```
results/metrics/deforestation_metrics.csv
```

Columns include:

- Municipality ID and name.
- Year and source (MapBiomas or PRODES).
- Total deforested area (ha).
- Patch count, mean and median patch area (ha).
- Entropy and mutual information metrics.

All scripts, configuration files, and logs are tracked in a controlled repository for reproducibility.

Appendix H. Land use conversions during 2004 to 2012

Table H.1 presents the transitions from forest to various anthropogenic land use classes in Brazilian Amazon municipalities between 2004 and 2012, using data from MapBiomas Collection 9. The values are disaggregated by two time periods–2004–2008 and 2008–2012–and separately for municipalities that were blacklisted under Brazil's environmental enforcement policy and those that were not. For each class, the table shows the total area of deforestation (in thousands of hectares) and its percentage share of total forest loss. Pasture dominates as the main post-deforestation land use across all groups and periods, especially in blacklisted municipalities where it accounts for nearly all deforestation. Agri-pasture mosaics are the second most common transition, followed by

Table H.1
Forest (3) → Anthropogenic Land Use Transitions (000 ha and %) by Municipality Type and Period.

		Non-blacklisted				Blacklisted			
		2004–2008		2008–2012		2004–2008		2008–2012	
Code	Land Use Class	Area	%	Area	%	Area	%	Area	%
15	Pasture	4817.70	79.66 %	2403.87	69.04 %	2743.34	98.95 %	1304.89	98.02 %
21	Agri-Pasture Mosaic	962.73	15.92 %	822.08	23.61 %	12.48	0.45 %	8.70	0.65 %
9	Forest Plantation	150.12	2.48 %	168.14	4.83 %	5.23	0.19 %	2.80	0.21 %
41	Other Temp Crops	45.91	0.76 %	35.10	1.01 %	7.32	0.26 %	7.46	0.56 %
39	Soybean	21.15	0.35 %	7.31	0.21 %	0.43	0.02 %	0.20	0.01 %
48	Other Perennial Crops	14.54	0.24 %	8.13	0.23 %	–	–	–	–
30	Mining	5.75	0.10 %	8.30	0.24 %	1.98	0.07 %	5.83	0.44 %
20	Sugar cane	4.73	0.08 %	3.79	0.11 %	–	–	–	–
47	Citrus	4.13	0.07 %	4.06	0.12 %	–	–	–	–
46	Coffee	4.06	0.07 %	8.31	0.24 %	–	–	–	–
24	Urban Area	3.59	0.06 %	5.89	0.17 %	0.18	0.01 %	0.23	0.02 %
35	Palm Oil	1.56	0.03 %	0.61	0.02 %	0.26	0.01 %	0.22	0.02 %
40	Rice	0.49	0.01 %	0.28	0.01 %	–	–	–	–
62	Cotton (beta)	≈0.00	≈0.00 %	0.00	0.00 %	0.00	0.00 %	0.00	0.00 %
25	Other Non-Vegetated	11.42	0.19 %	5.78	0.17 %	1.31	0.05 %	0.89	0.07 %

Notes: This table reports the total area (in 000 hectares) and percentage share of forest loss that transitioned to anthropogenic land uses in Brazilian Amazon municipalities, using MapBiomas Collection 9 transition data. Values are separately reported for blacklisted and non-blacklisted municipalities over two periods: 2004–2008 and 2008–2012. Dashes (–) indicate zero or negligible conversions for that class-period-municipality combination. Class codes correspond to MapBiomas standard categories. Pasture (15) and Agri-Pasture Mosaic (21) dominate post-deforestation transitions, particularly in blacklisted areas.

smaller contributions from forest plantations, temporary crops, and other uses. The table highlights distinct land use patterns following deforestation, with blacklisted municipalities showing more concentrated transitions to pasture, while non-blacklisted areas exhibit slightly more diversified land conversions.

Data availability

Data will be made available upon request.

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